

Graphical expression

Solid Edge project

Humanoid robot

[redacted]

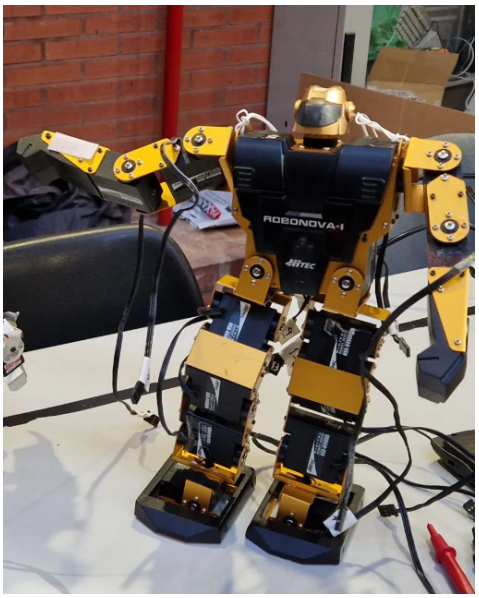
[redacted]

Iris Valverde Blanco [redacted]

[redacted]

Introduction:

For this project we have modeled in Solid Edge a mini humanoid robot, as the one shown in the picture.



To simplify the construction of the robot, we have distinguished four main parts: the torso, the legs, the arms and the motors.

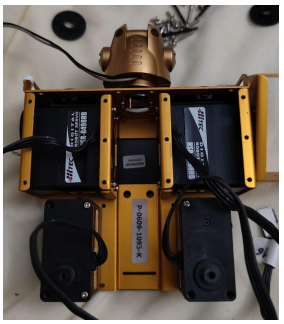
- There are three types of motors: a single motor with only one axis that is used for the shoulders, a double motor and a motor with protrusions used in both legs and arms.



- The arm consists of a hand, a wrist, two motors (double and one with protrusions), two arrows and one U-piece to join with the torso (plus screws). (U-piece missing in the foto)



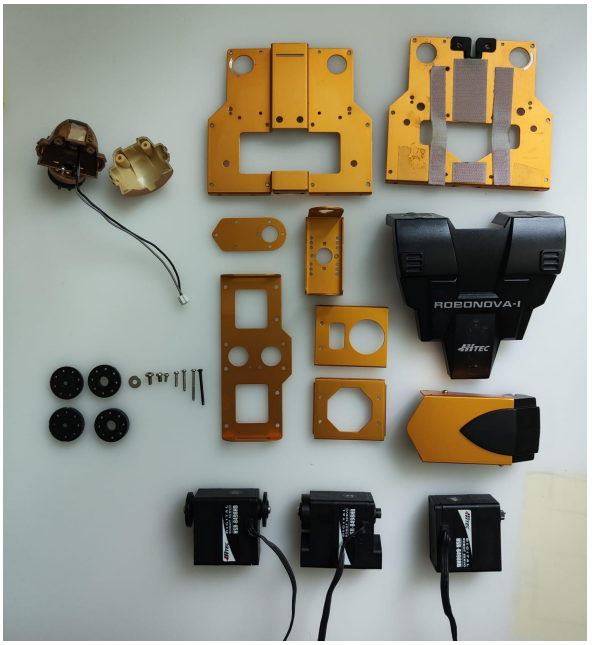
- The torso has: two double motors, two single motors, two ribs, two armpits, a front piece, a back piece, a top piece, a head and a chest (plus screws).





- The leg is made of four U-pieces, four motors (three with protrusions and one double), a double arrow, two single arrows, a base and a foot (plus screws).

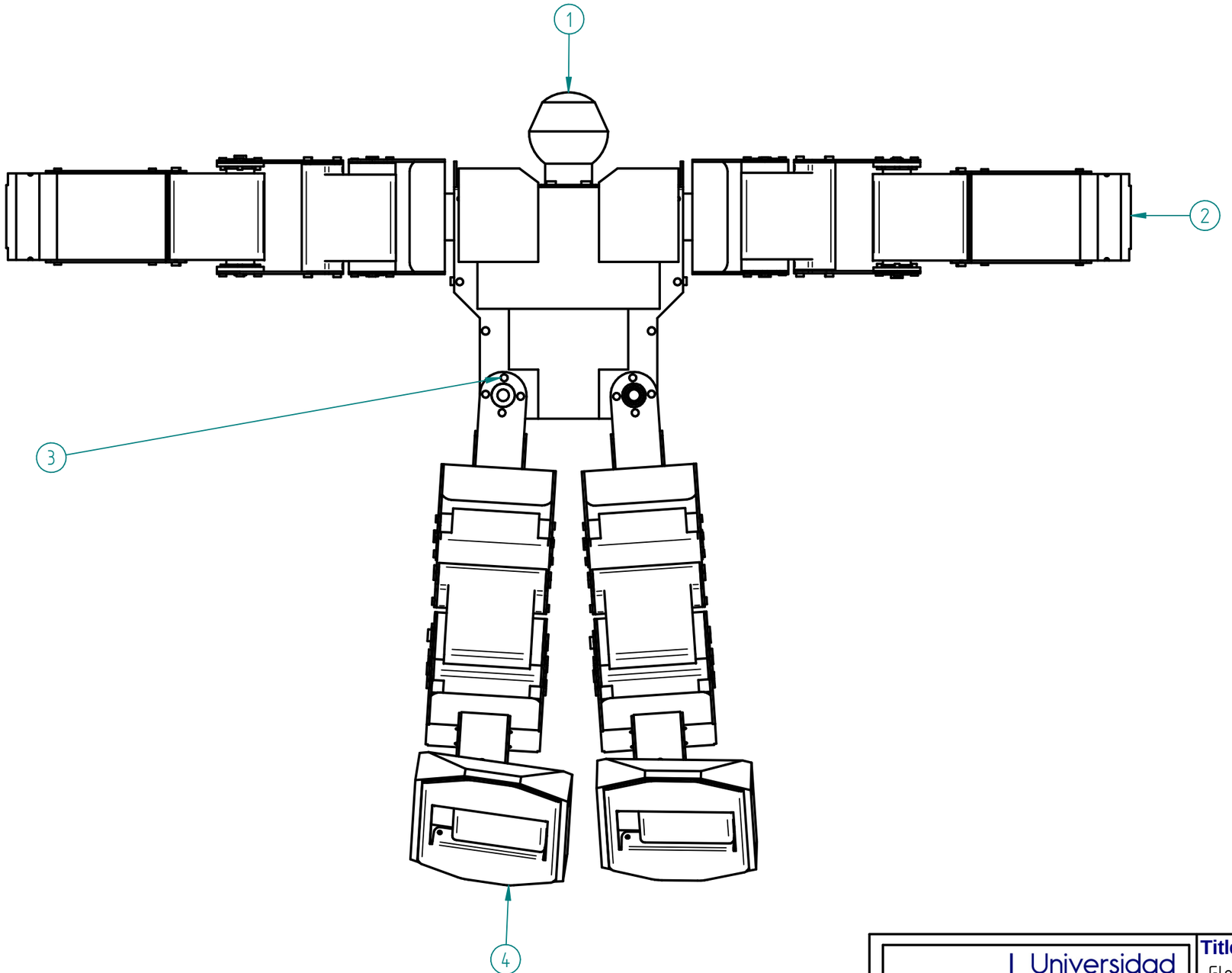




(most of the pieces individually)

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Item Number	Title	Quantity
4	Leg - subassembly	2
3	Screw2	24
2	Arm - subassembly	2
1	Torso - subassembly	1

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Created by: [redacted]

Title: Global draft

Complementary title:

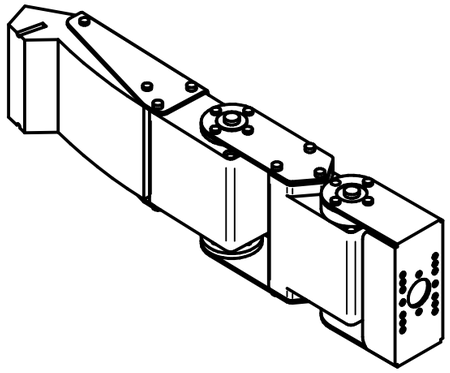
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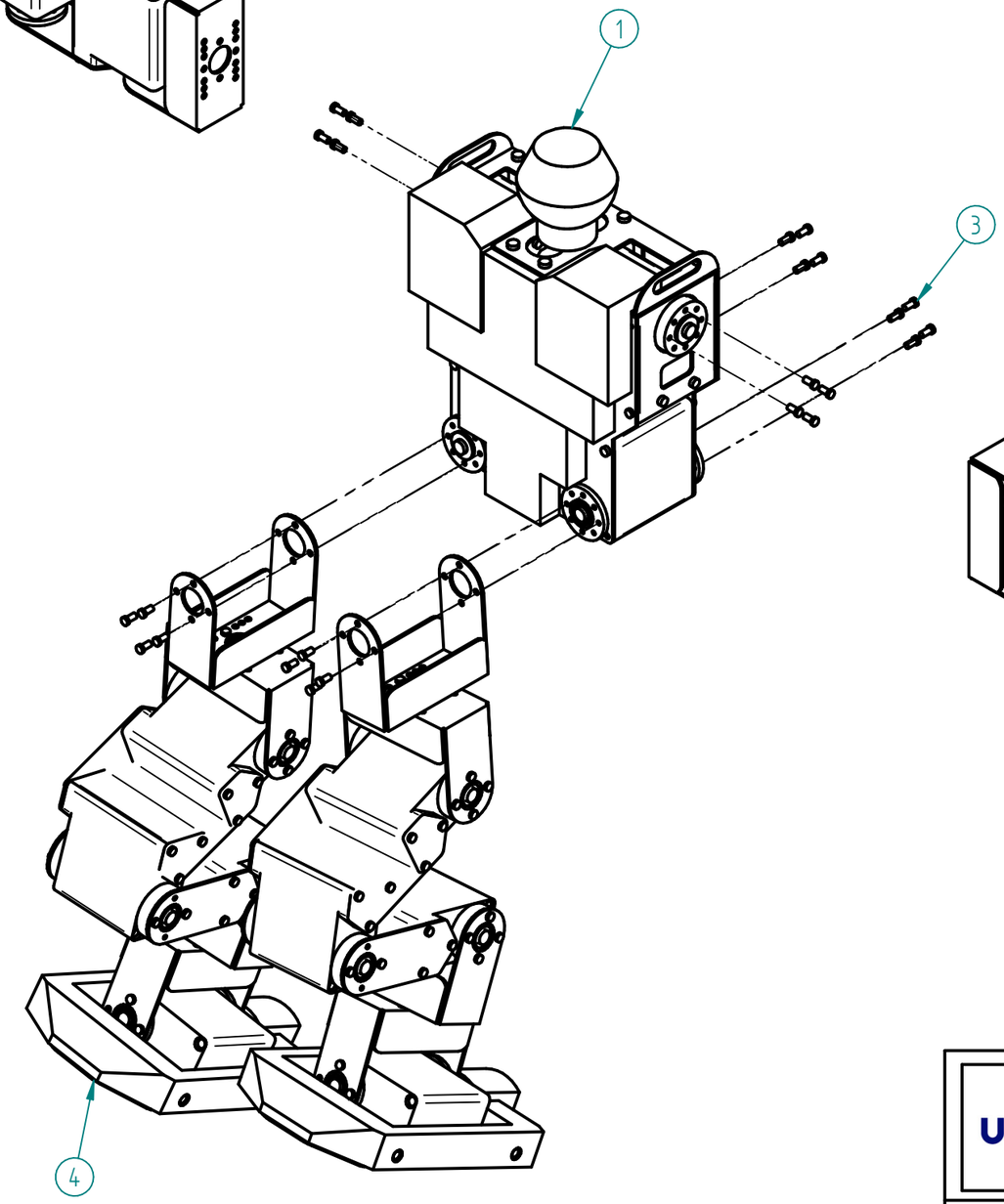
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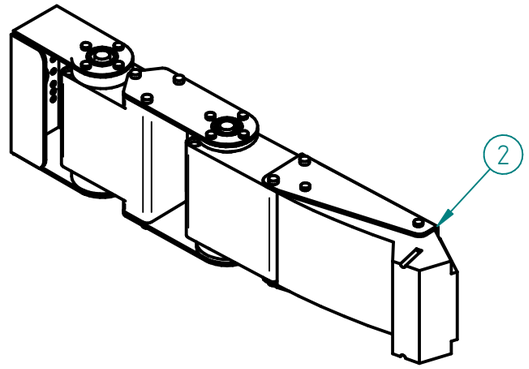
Date: 10/12/2023

Page number:

SOLID EDGE ACADEMIC COPY







Item Number	Title	Quantity
4	Leg – subassembly	2
3	Screw2	24
2	Arm – subassembly	2
1	Torso – subassembly	1

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Created by: [redacted]

Title: Global exploded

Complementary title:

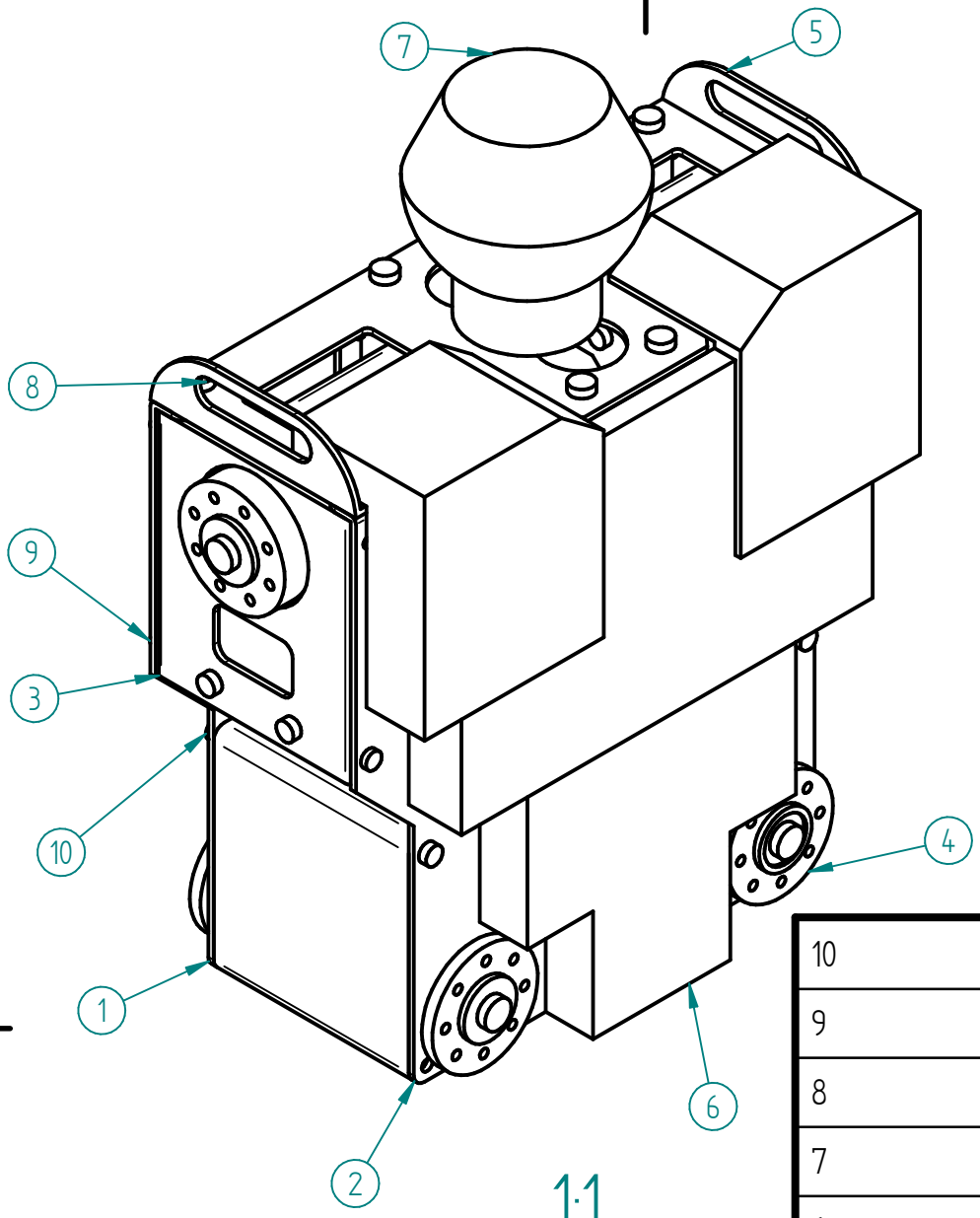
Kind of draft: Global assembly exploded draft

Scale: 1:2

Date: 10/12/2023

Page number:

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1:1

Item Number	Title	Quantity
10	Screw1	4
9	Screw2	16
8	Screw3	10
7	Head	1
6	chest	1
5	Top	1
4	Double Motor	2
3	Rib Motor	2
2	Front	1
1	Back	1

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Created by:
Iris Valverde Blanco

Title:
Torso subassembly

Complementary title:

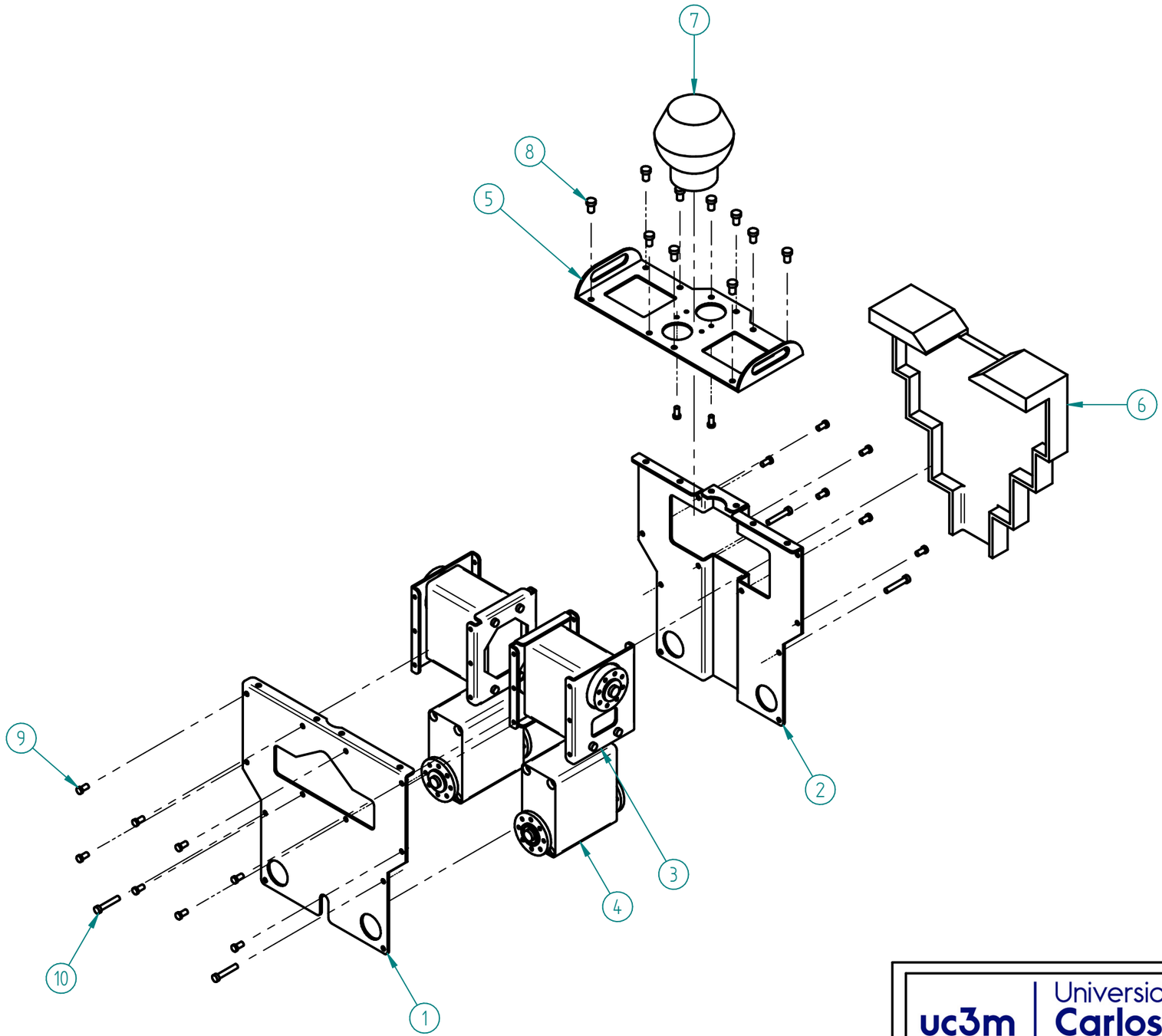
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Subassembly draft

Scale:
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Date:
10/12/2023

Page number:

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Item Number	Title	Quantity
10	Screw1	4
9	Screw2	16
8	Screw3	10
7	Head	1
6	chest	1
5	Top	1
4	Double Motor	2
3	Rib Motor	2
2	Front	1
1	Back	1

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Created by:
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Title:
Exploded torso subassembly

Complementary title:

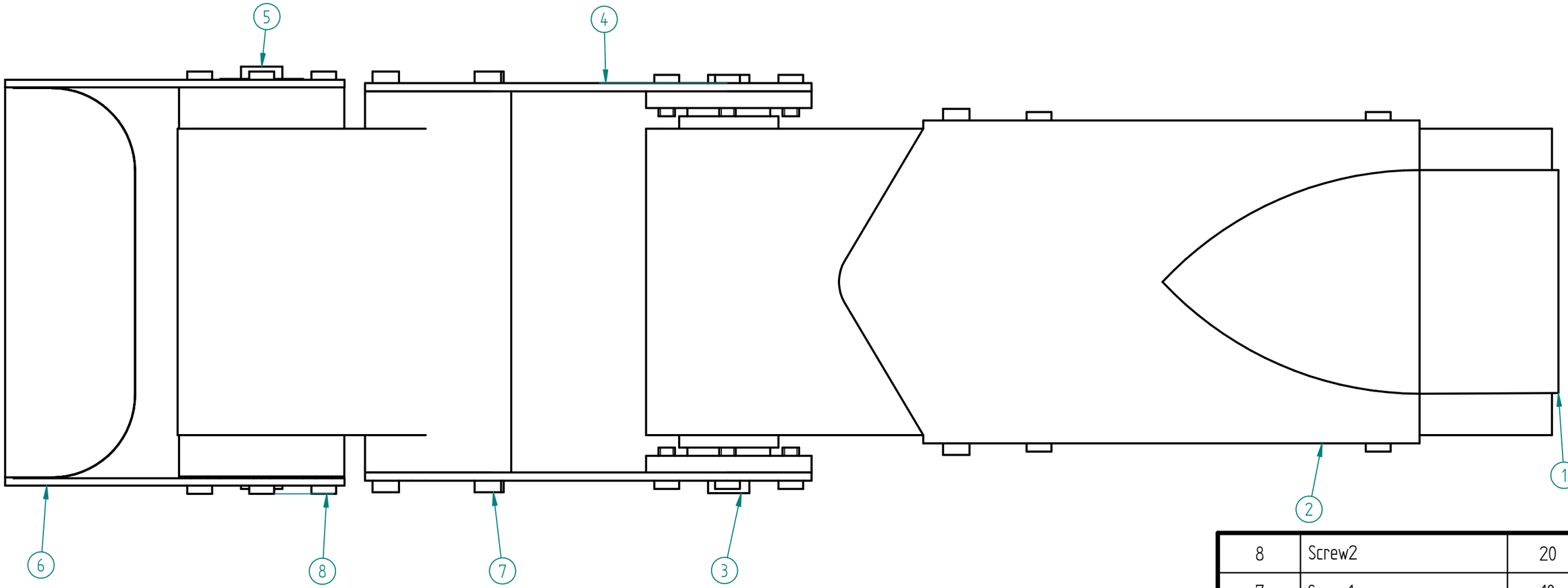
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Exploded subassembly draft

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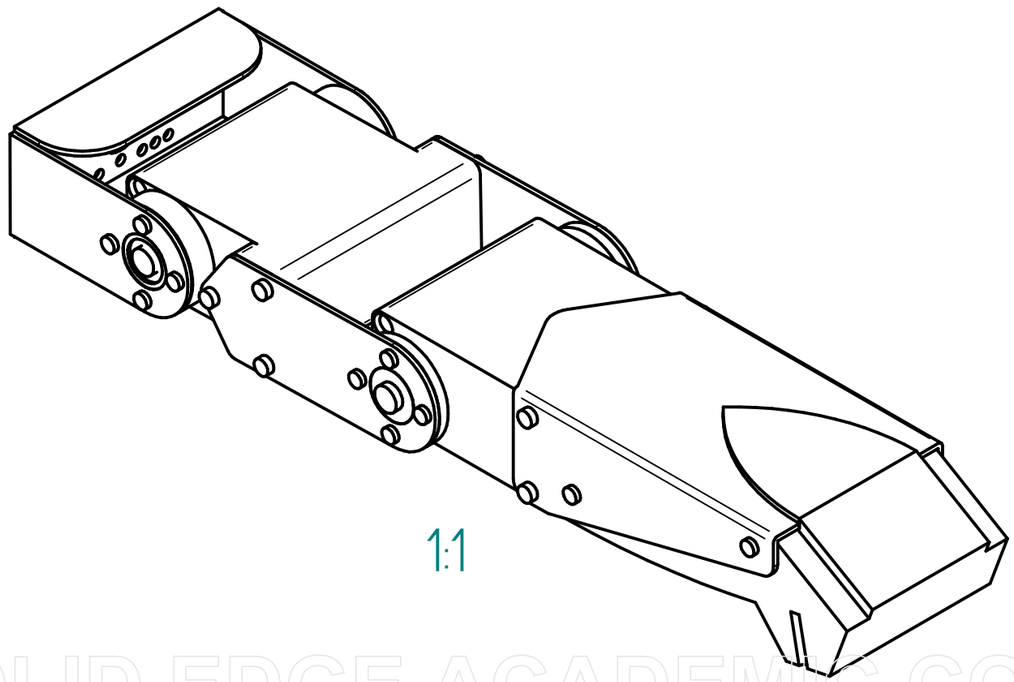
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10/12/2023

Page number:

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Item Number	Title	Quantity
8	Screw2	20
7	Screw1	10
6	U-part	1
5	Motor_with_protusions	1
4	Arrow	2
3	Double_motor	1
2	Wrist	1
1	Hand	1



1:1

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Created by: [redacted]

Title: Arm – subassembly

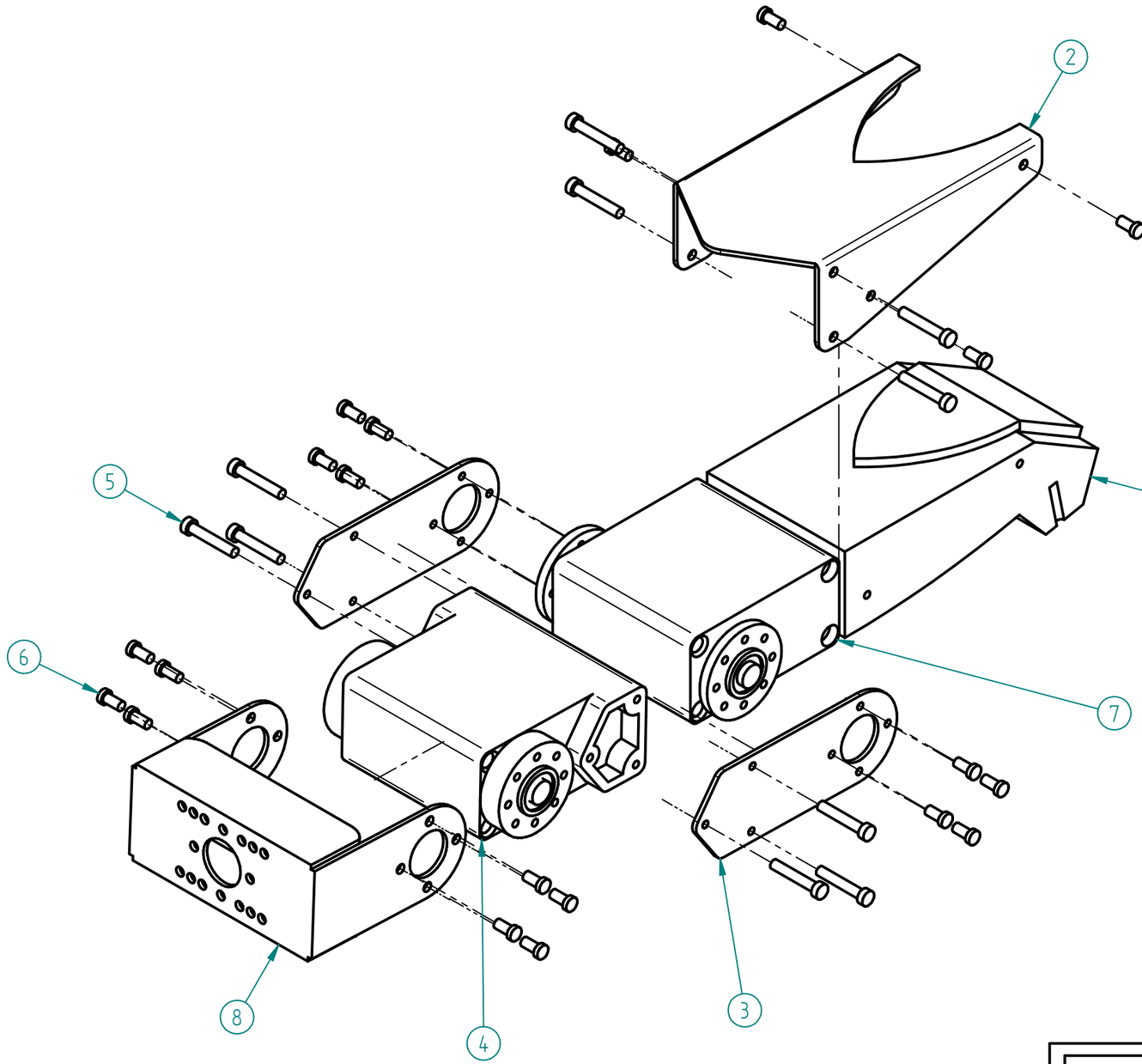
Complementary title:

Kind of draft: Subassembly draft

Scale: 2:1

Date: 9/12/2023

Page number:



Item Number	Title	Quantity
8	U-part	1
7	Double_motor	1
6	Screw2	20
5	Screw1	10
4	Motor_with_protusions	1
3	Arrow	2
2	Wrist	1
1	Hand	1

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Created by: [redacted]

Title: Exploded arm-subassembly

Complementary title:

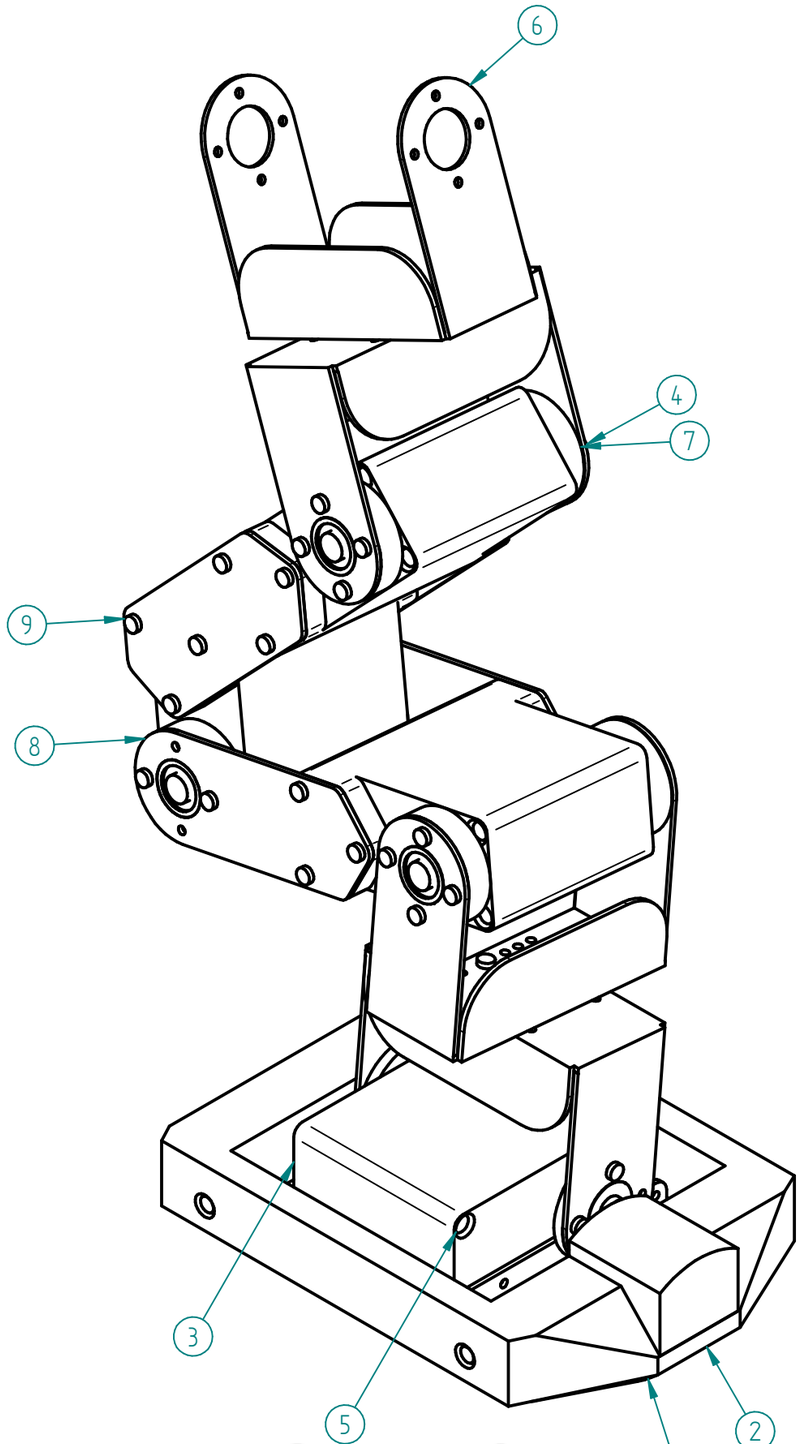
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Scale: 1:1

Date: 9/12/2023

Page number:

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Item Number	Title	Quantity
9	double arrow	1
8	arrow	2
7	motor with protusions	3
6	piezaU	4
5	screw1	2
4	screw2	52
3	double motor	1
2	foot	1
1	base	1

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Created by:

[redacted]

Title:
Leg - subassembly

Complementary title:

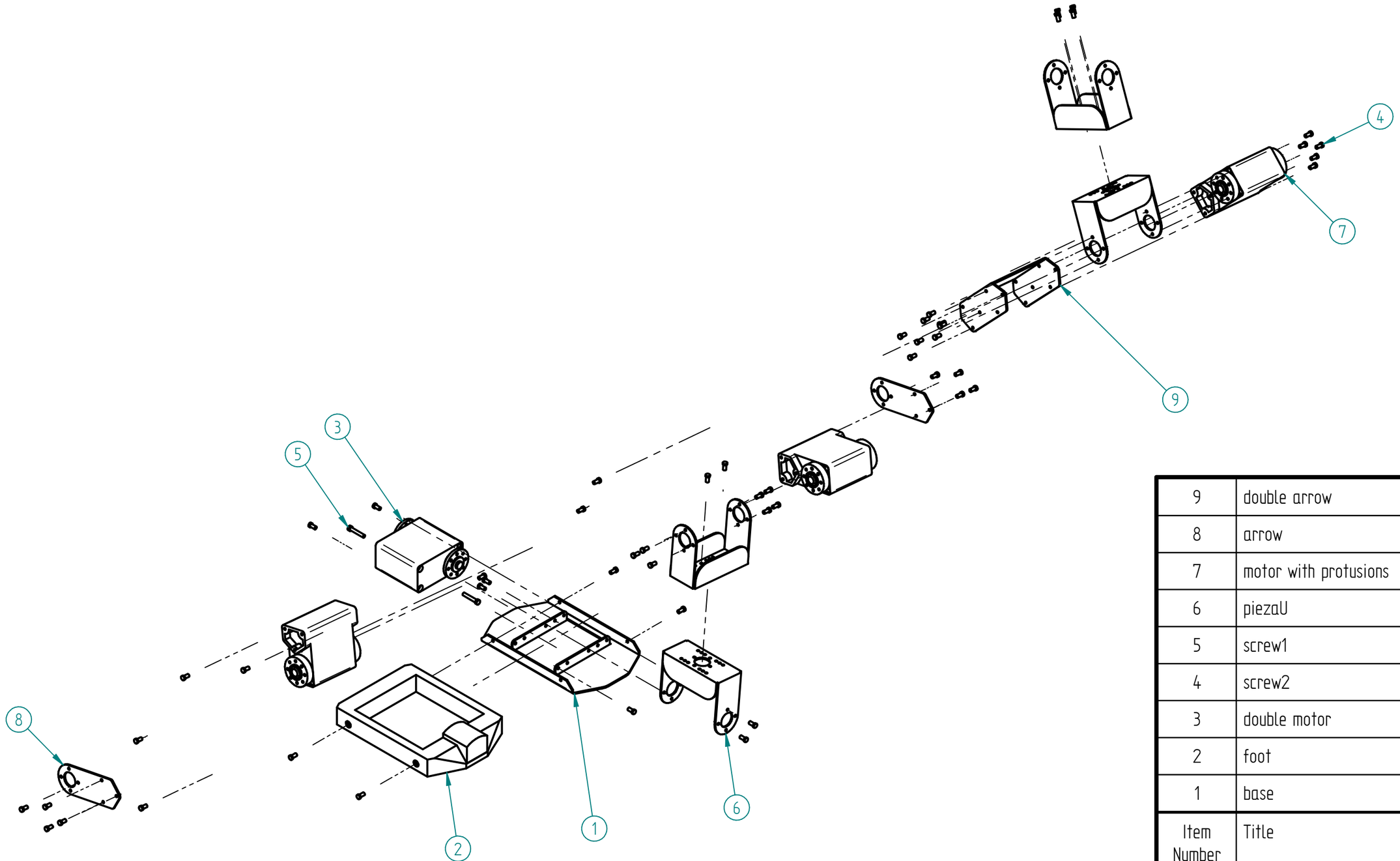
Kind of draft:
Subassembly draft

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9/12/23

Page number:

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Item Number	Title	Quantity
9	double arrow	1
8	arrow	2
7	motor with protusions	3
6	piezaU	4
5	screw1	2
4	screw2	52
3	double motor	1
2	foot	1
1	base	1

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Created by: [redacted]

Title: Exploded leg-subassembly

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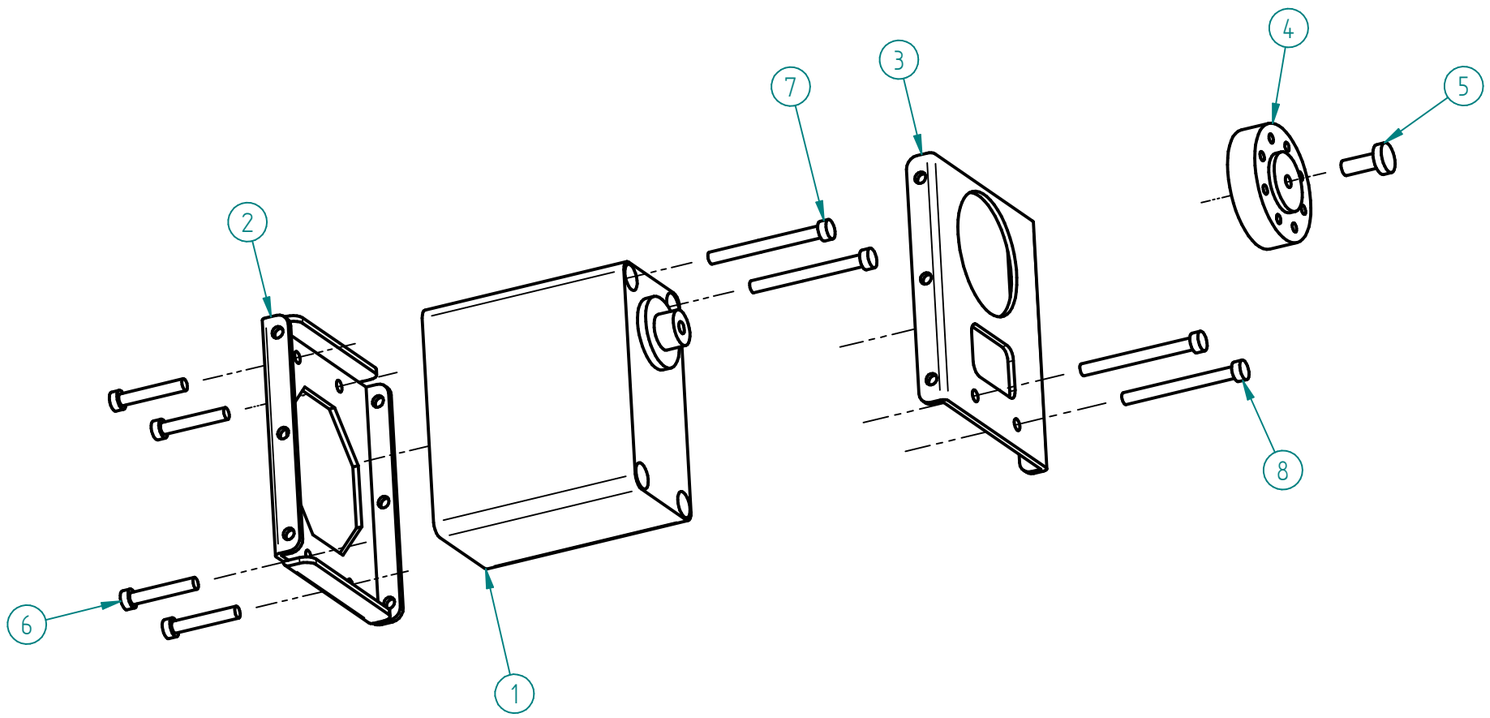
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Date: 9/12/23

Page number:

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Item Number	Title	Quantity
8	Screw 2.2 motor	2
7	Screw 2 motor	2
6	Screw 1	4
5	Screw motor axis	1
4	Thick horn axis	1
3	Armpit	1
2	Rib	1
1	Single motor	1

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Created by: [redacted]

Title: Rib motor exploded

Complementary title:

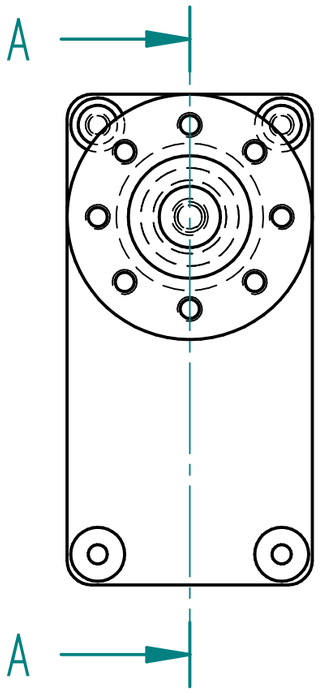
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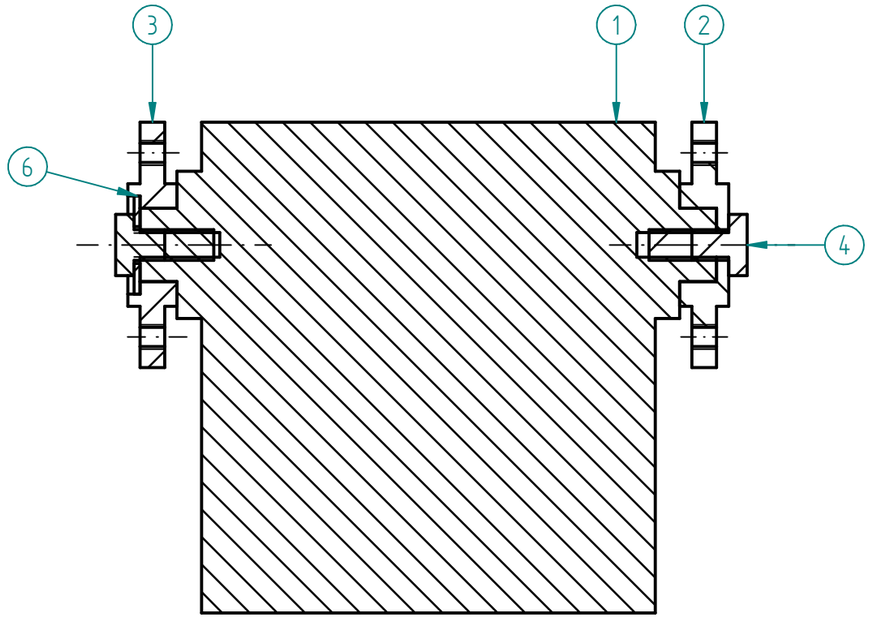
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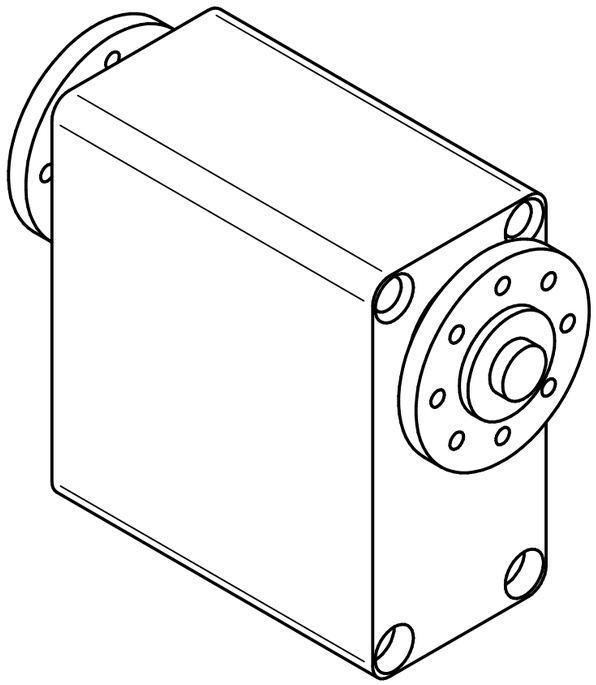
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CORTE A-A



Item Number	Title	Quantity
6	Washer	1
5*	Screw 1	4
4	Screw motor axis	2
3	Simple horn	1
2	Simple axis horn	1
1	Double motor	1

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Created by: [redacted]

Title: Double motor subassembly

Complementary title:

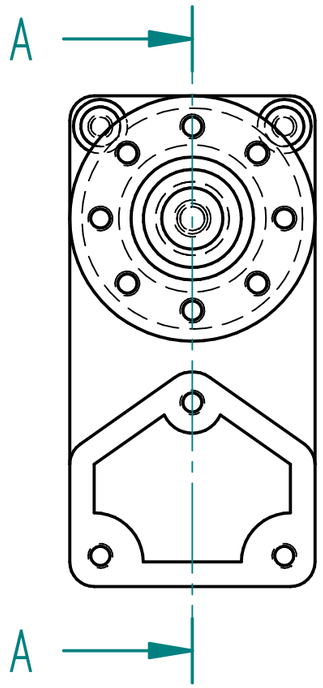
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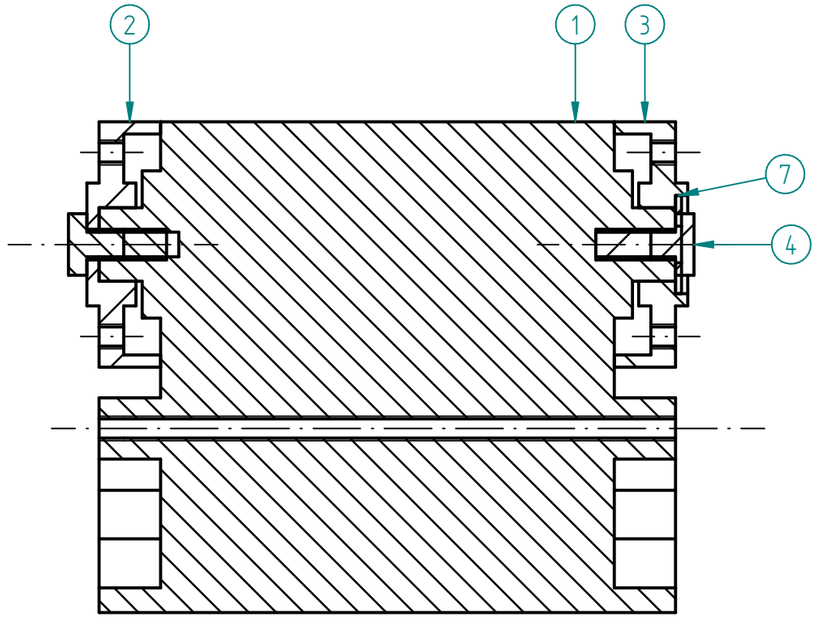
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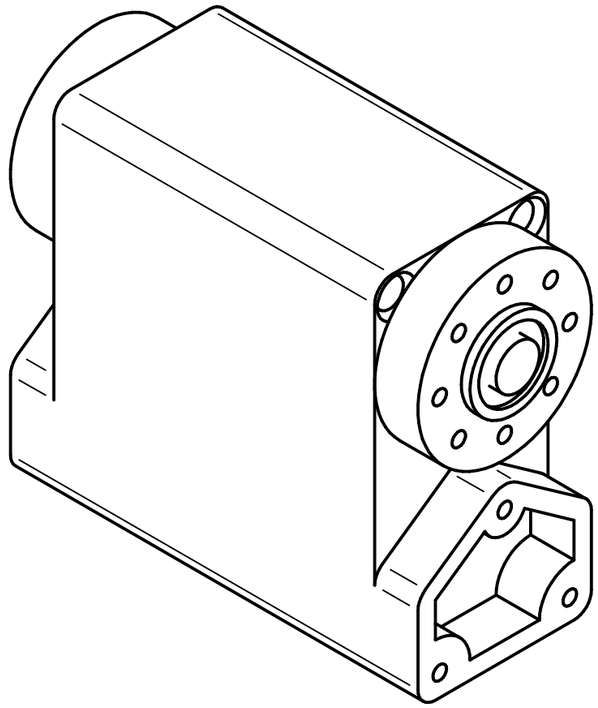
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CORTE A-A



Item Number	Title	Quantity
7	Washer	1
6*	Screw 2 motor	2
5*	Screw 1	2
4	Screw motor axis	2
3	Thick horn	1
2	Thick horn axis	1
1	Motor with protrusions	1

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Created by: [redacted]

Title: Motor with protrusions subassembly

Complementary title:

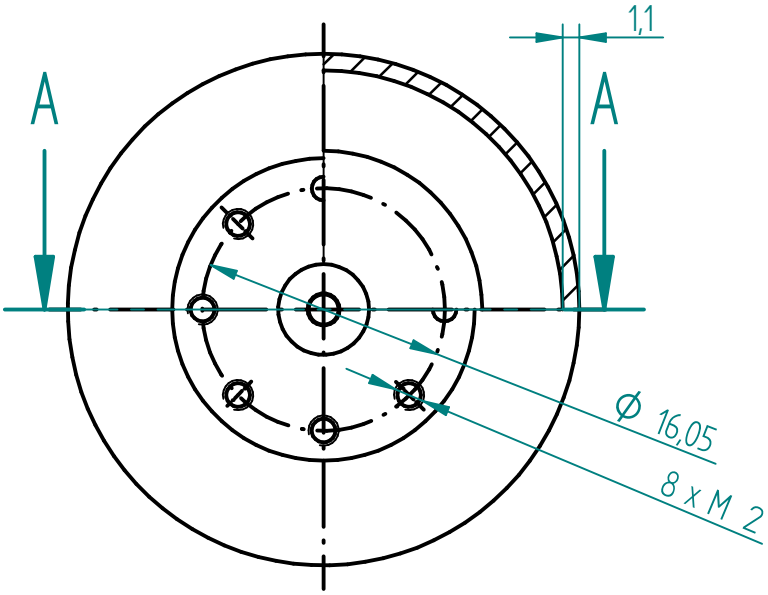
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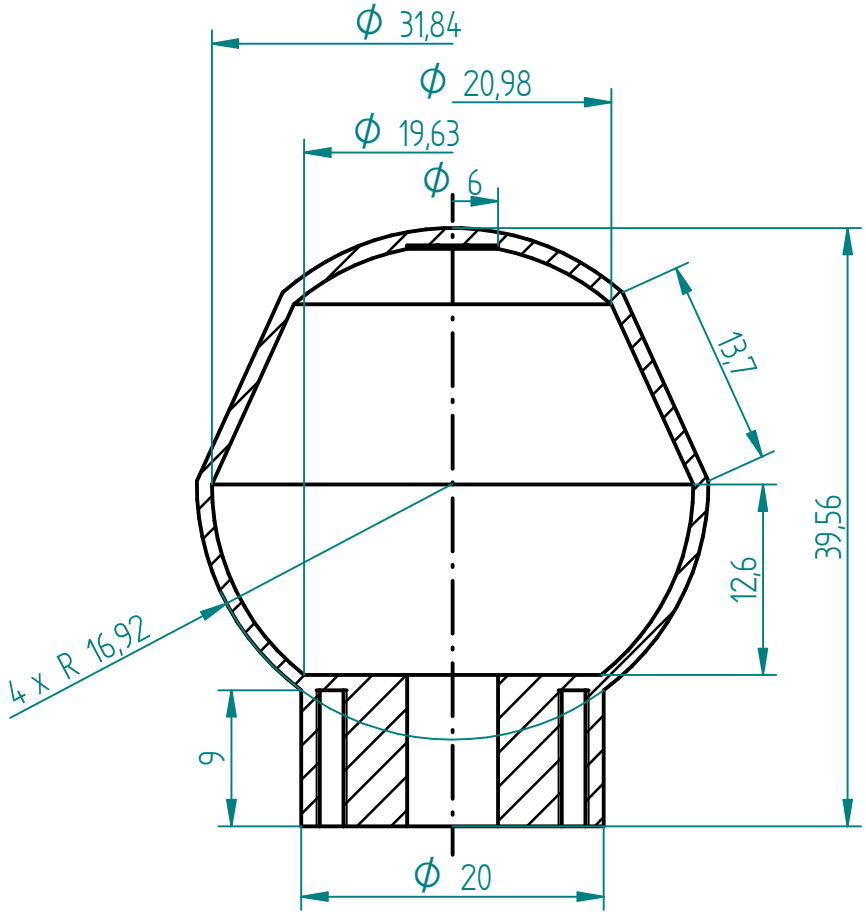
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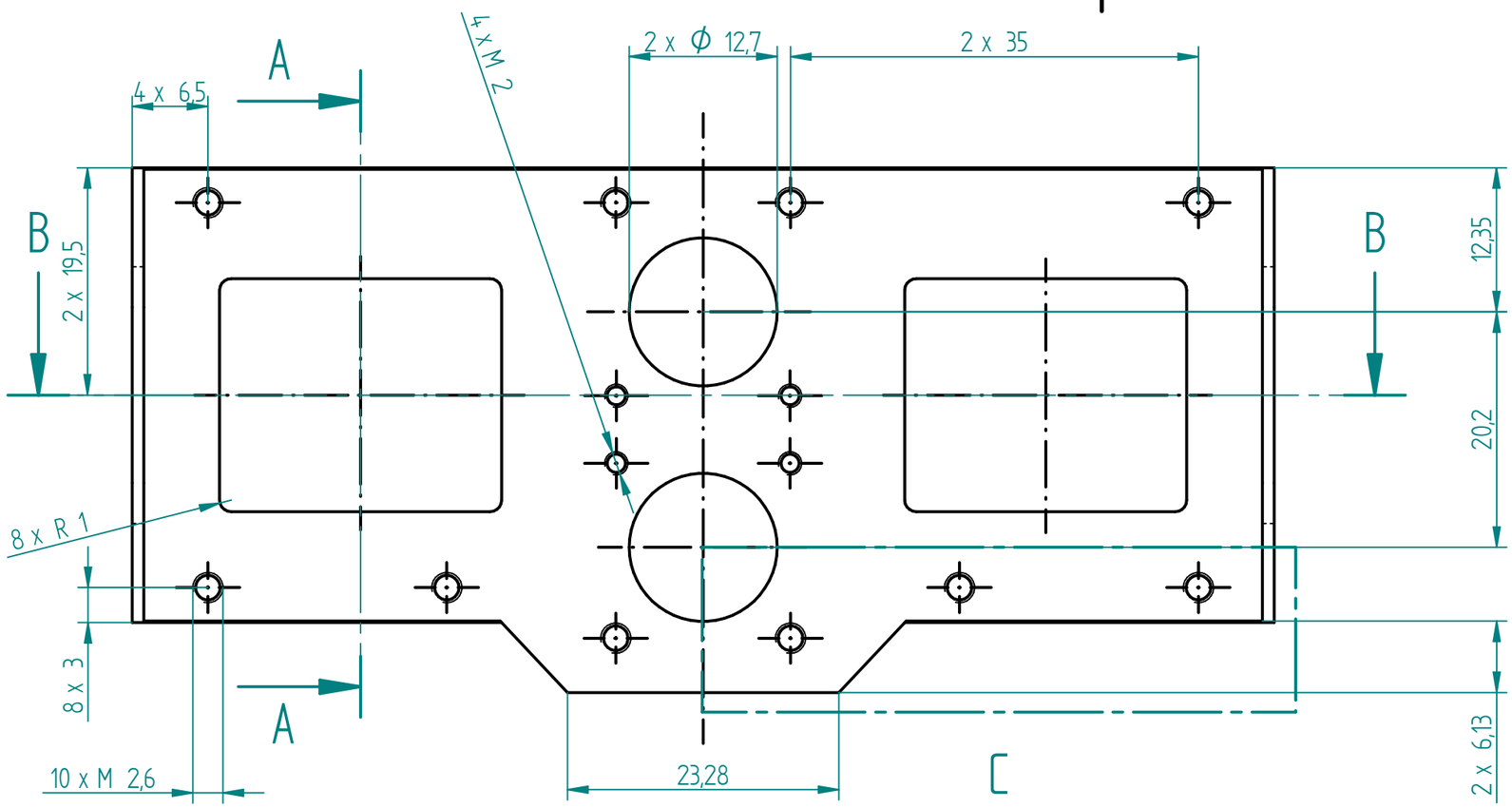


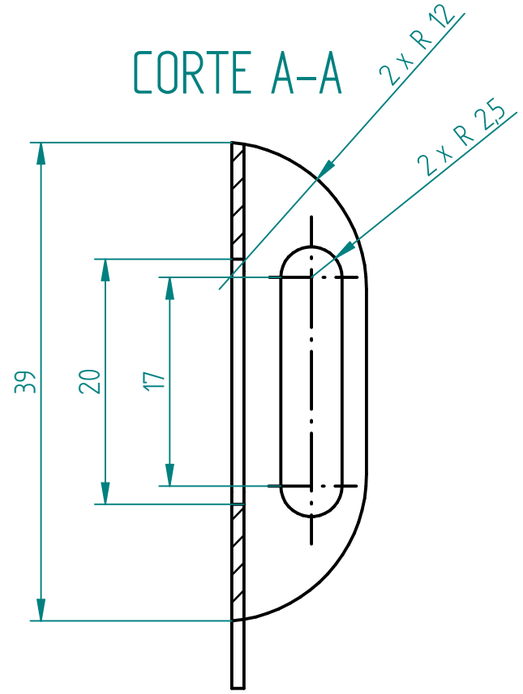


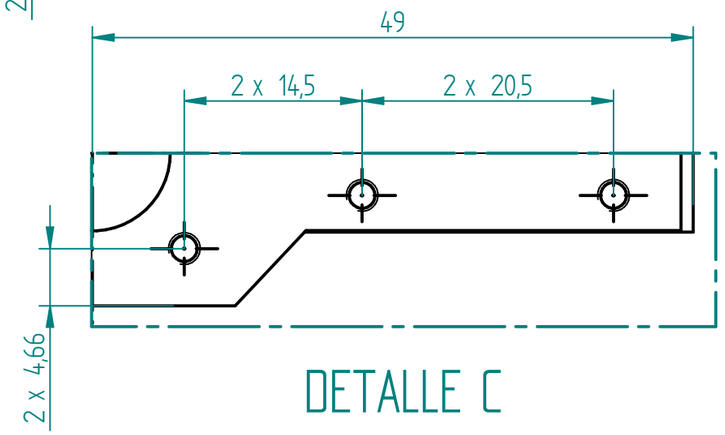
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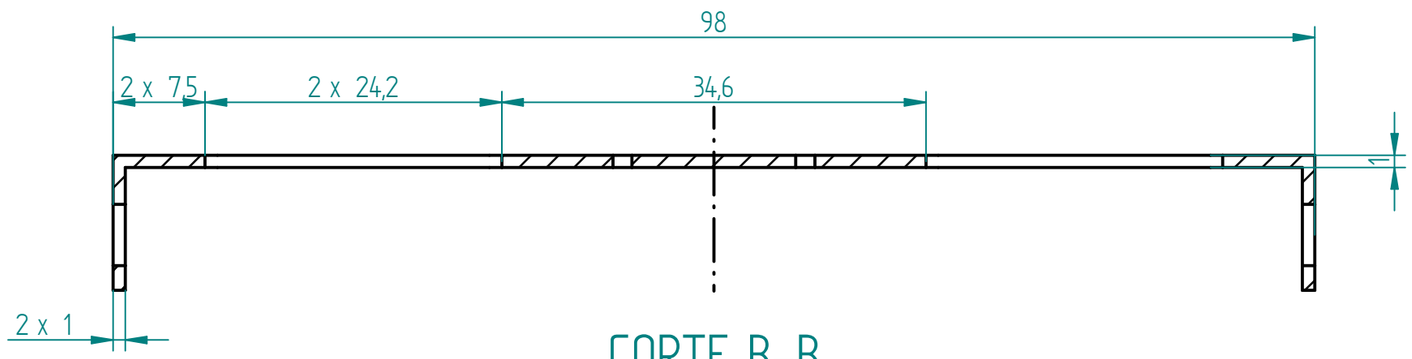
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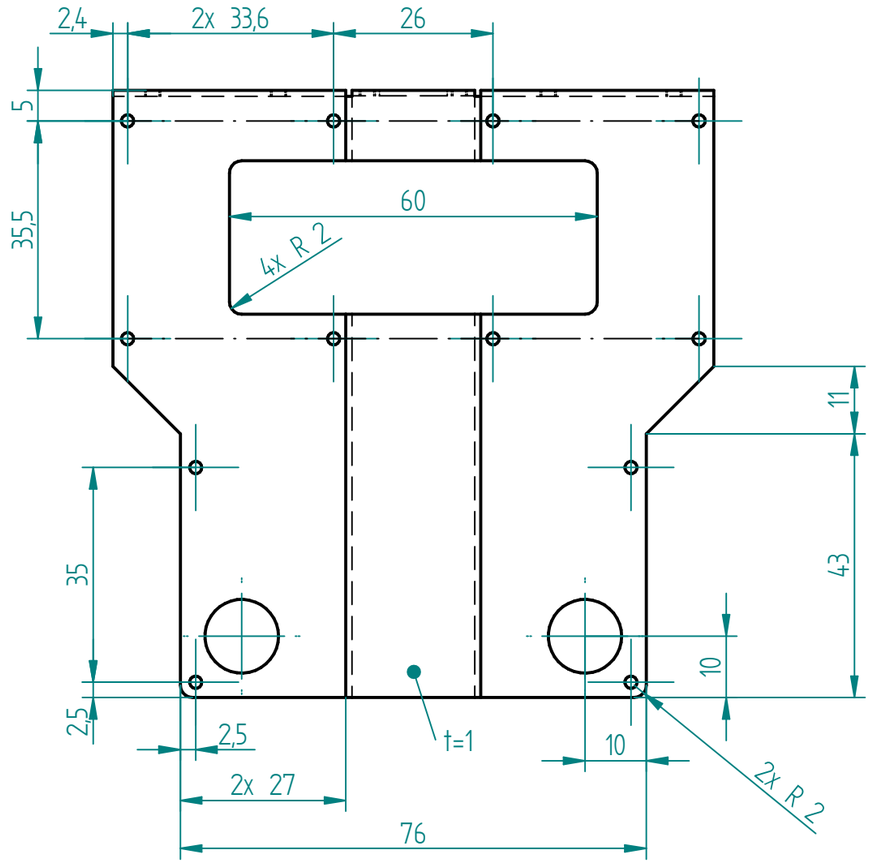
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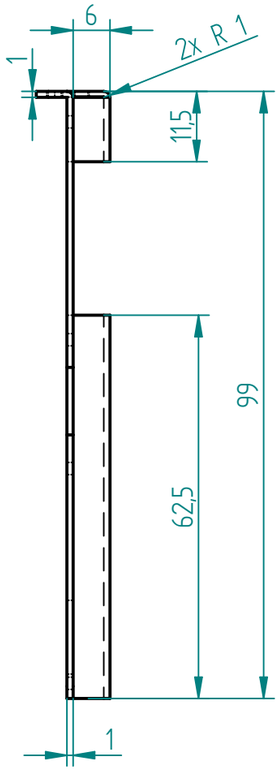


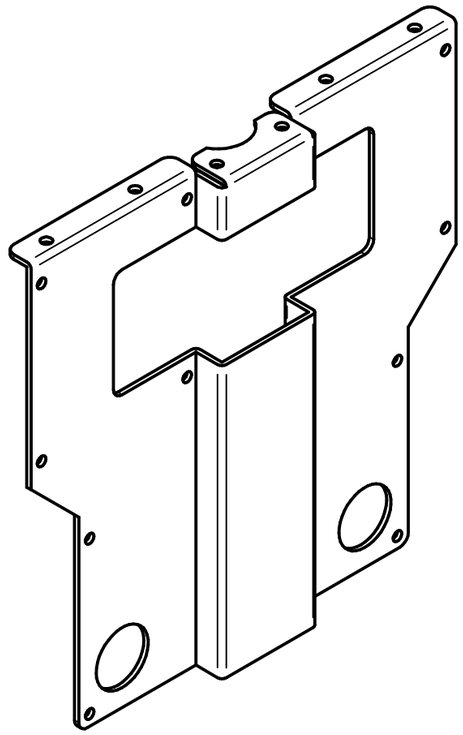
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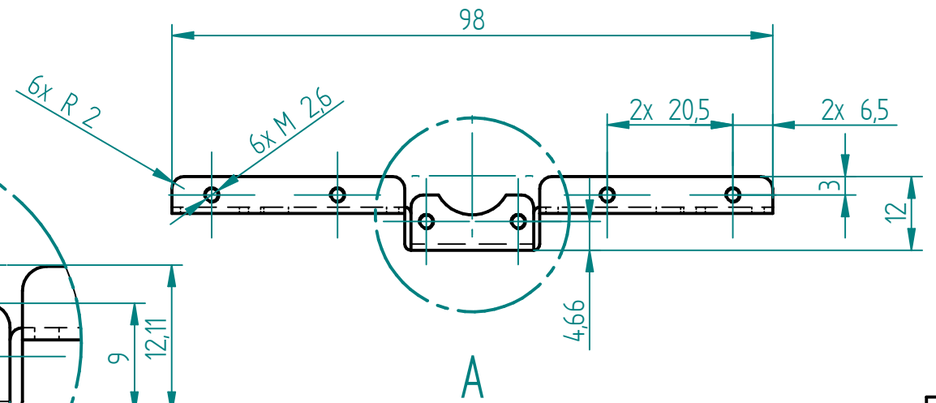
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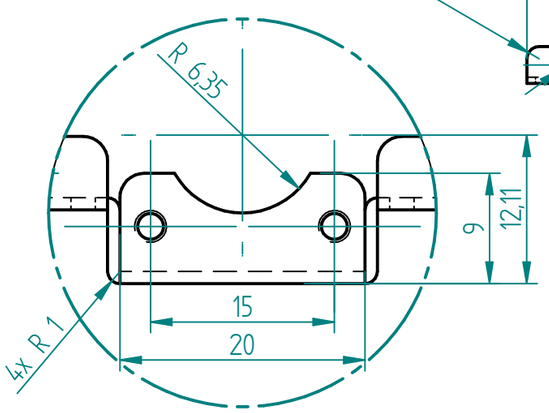
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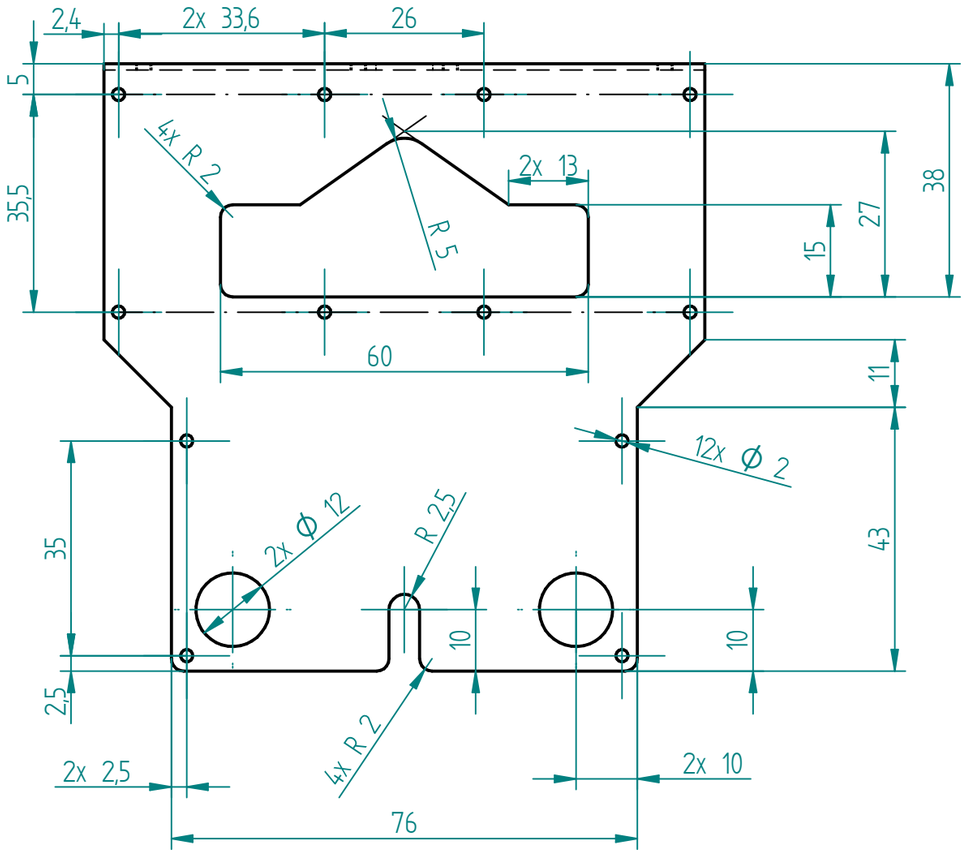


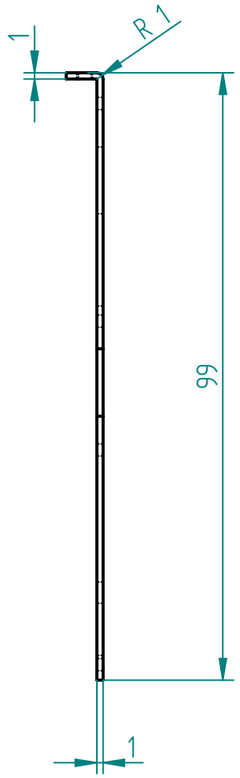


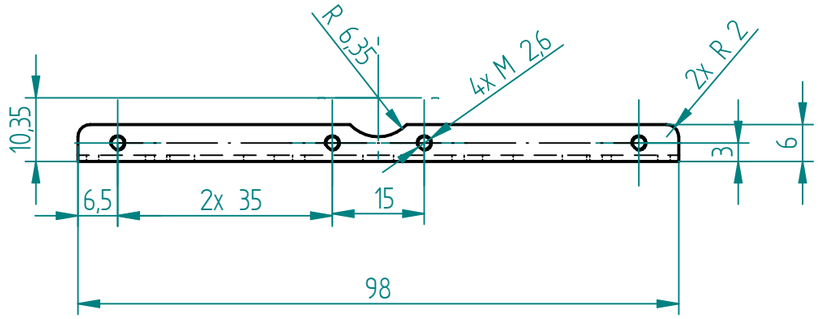
DETAIL A
2:1

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Created by: [redacted]	**Complementary title:**
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	Date: 3/12/2023
	Page number:







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Created by: [redacted]

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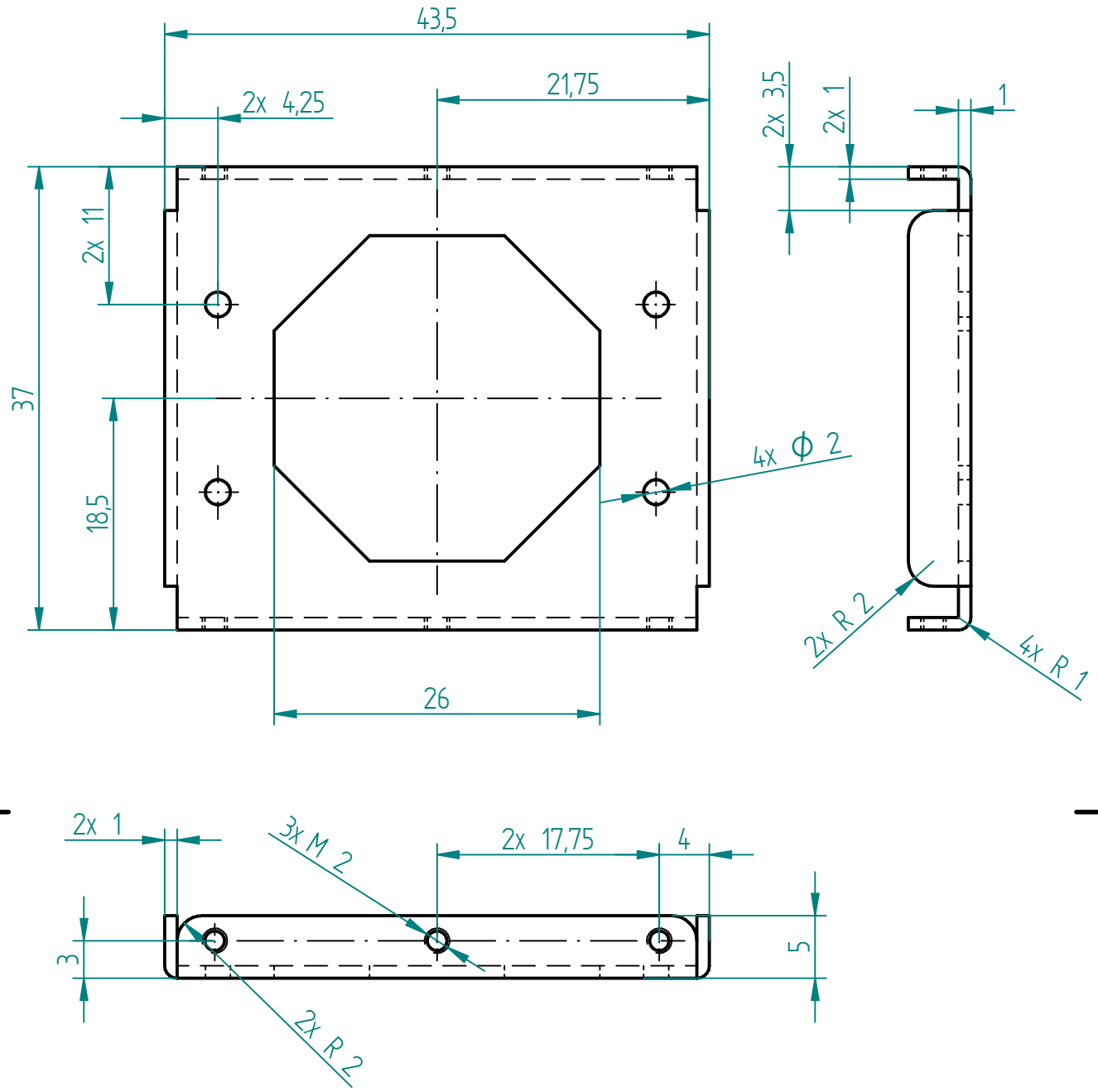
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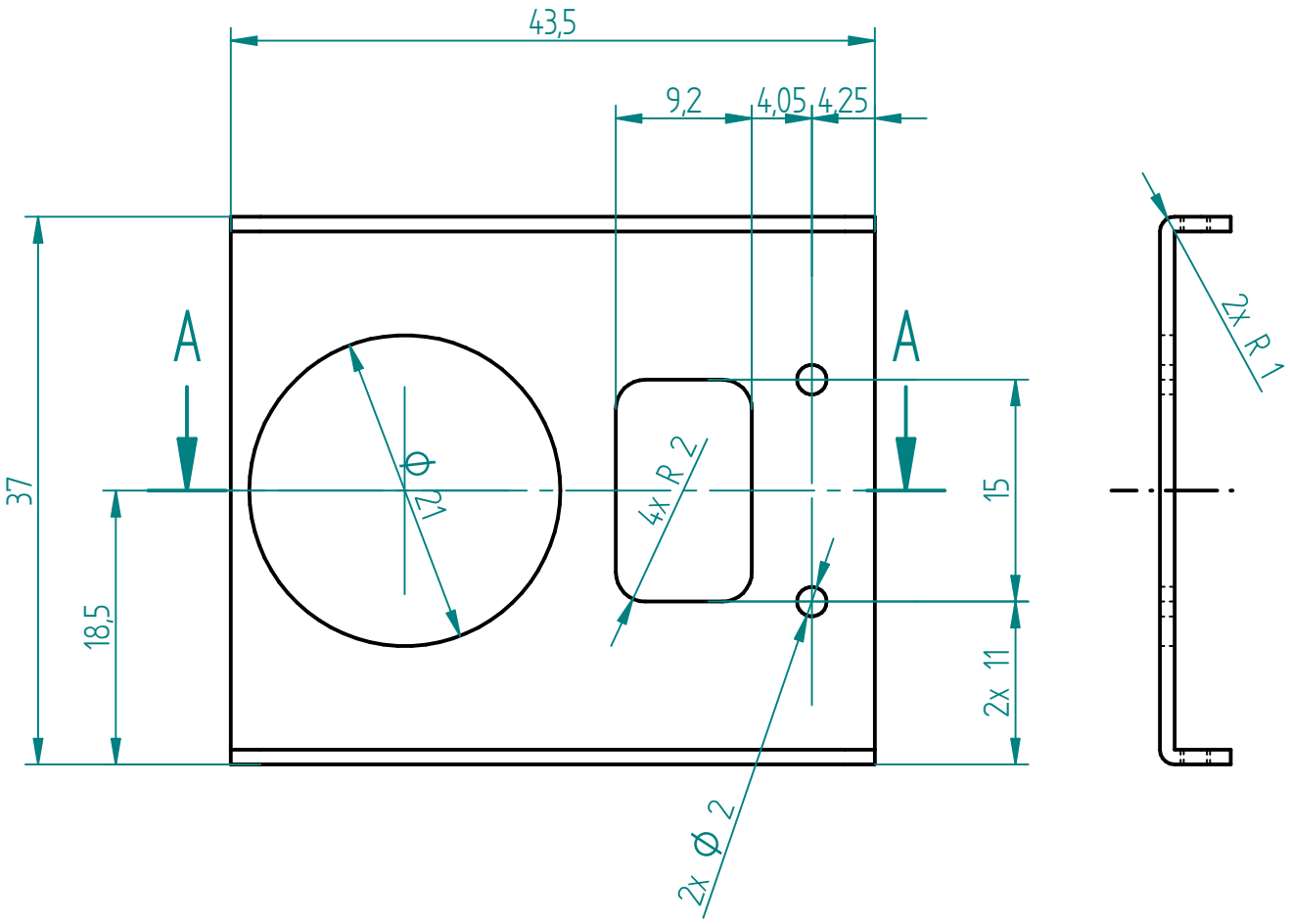
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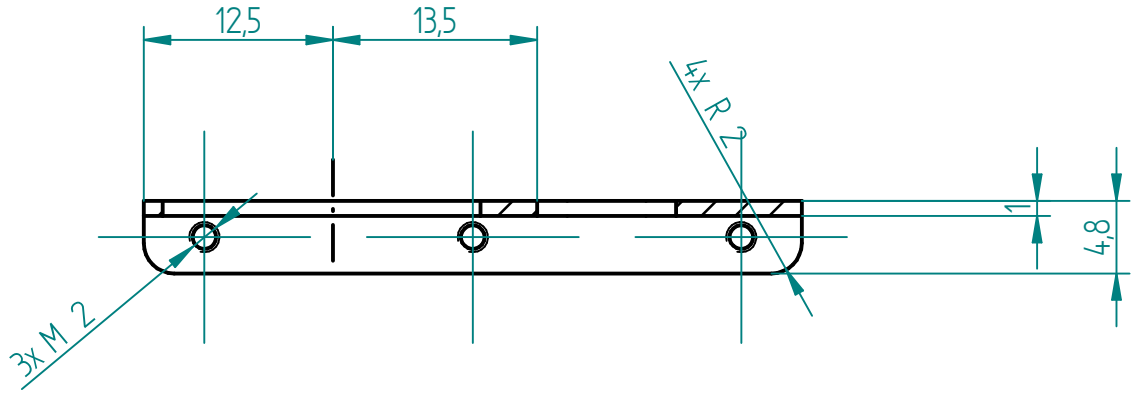
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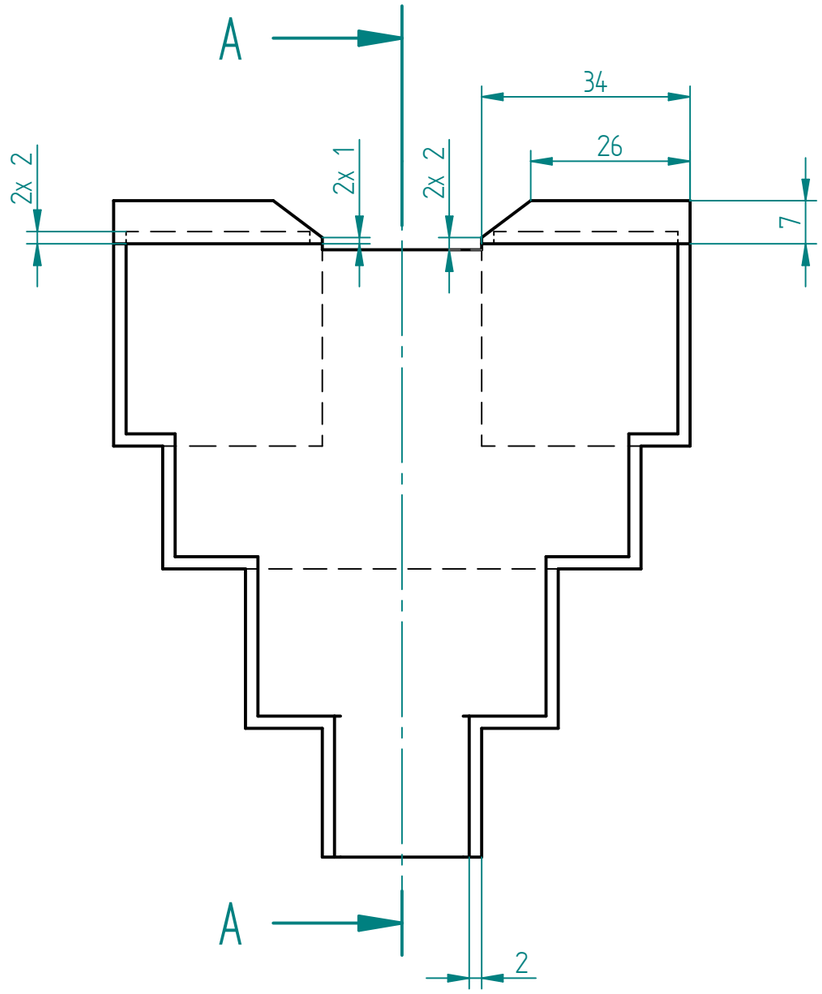


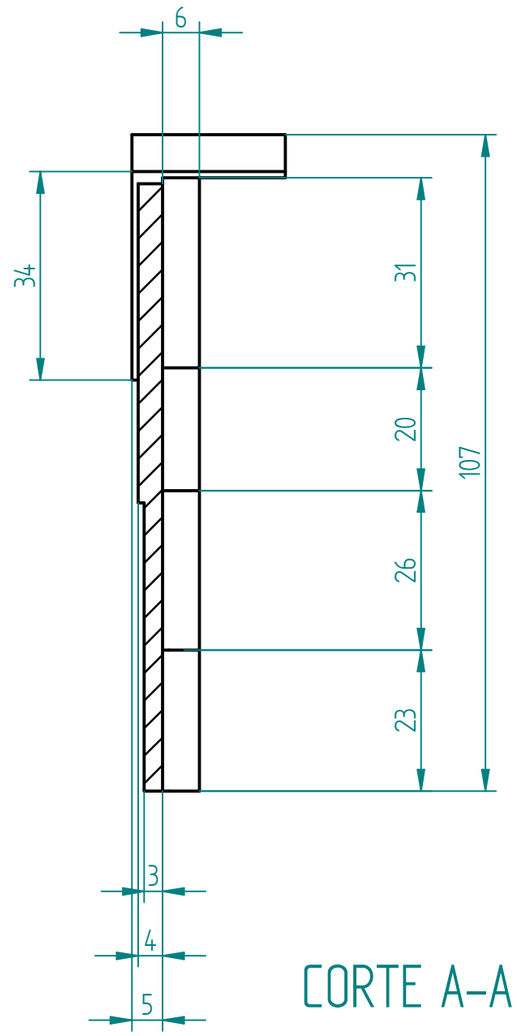


CORTE A-A

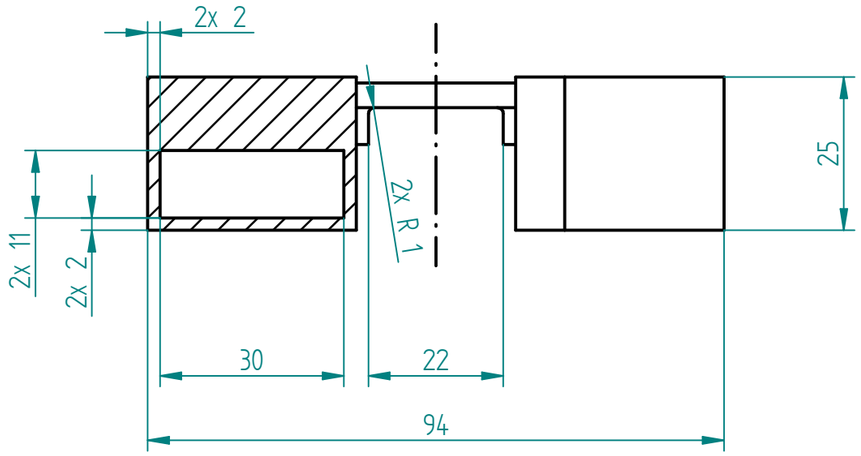
uc3m | Universidad Carlos III de Madrid	**Title:** Armpit	
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Created by: Iris Valverde Blanco		**Date:** 02/11/23
	Kind of draft: Part draft	**Page number:**

SOLID EDGE ACADEMIC COPY



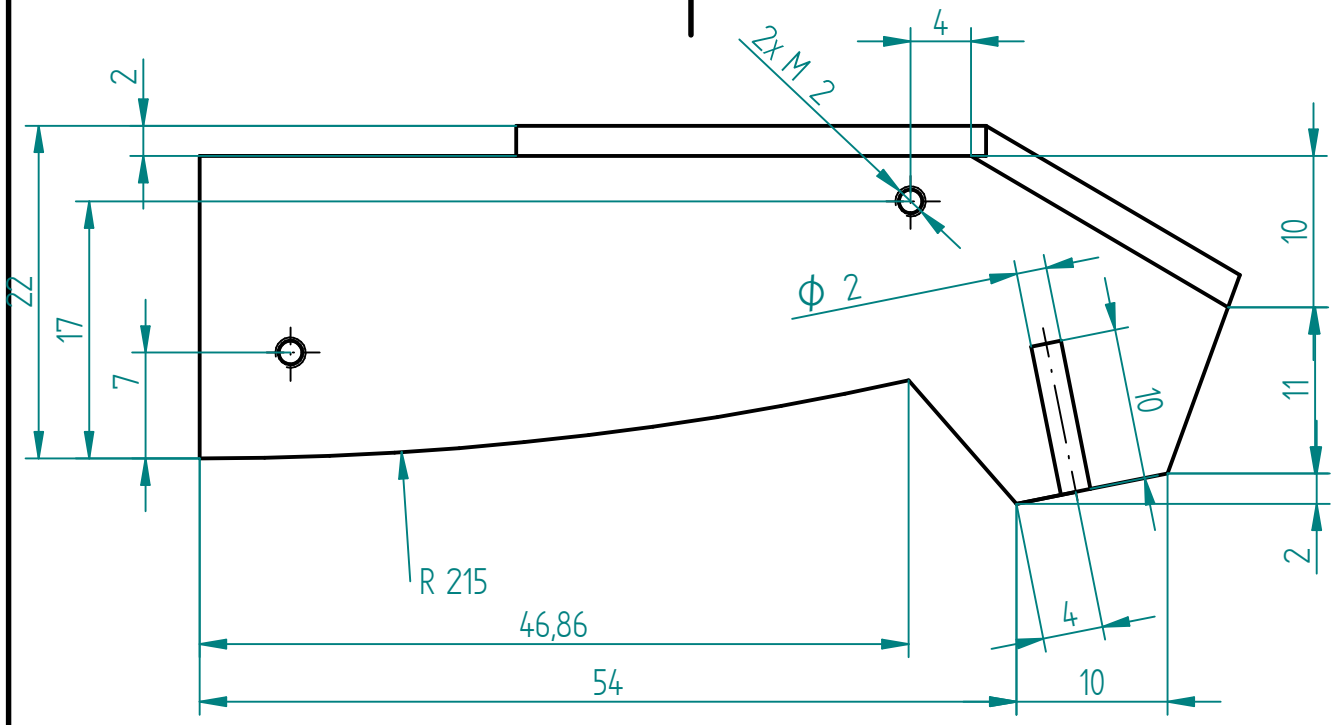


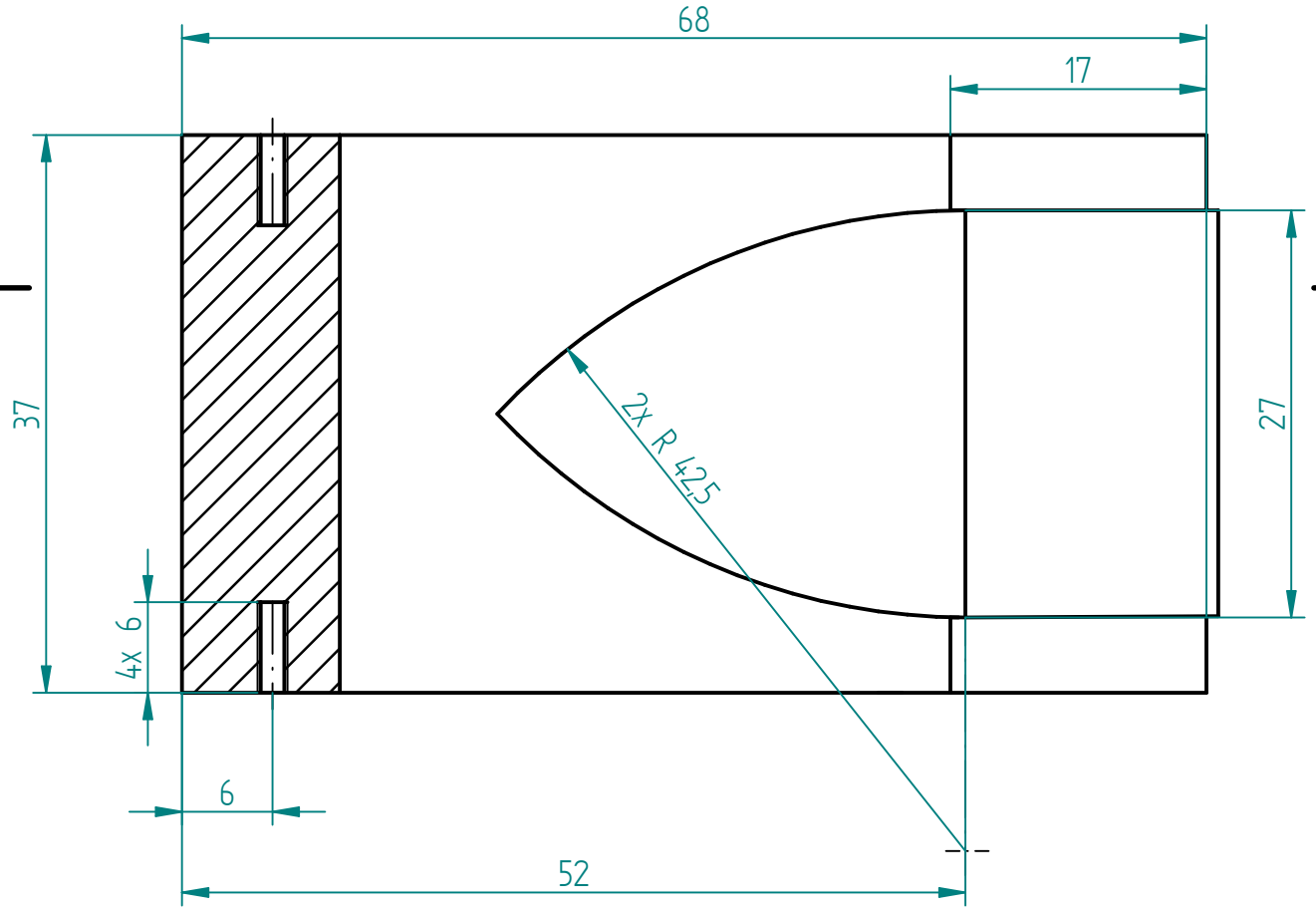
CORTE A-A



SOLID EDGE ACADEMIC COPY

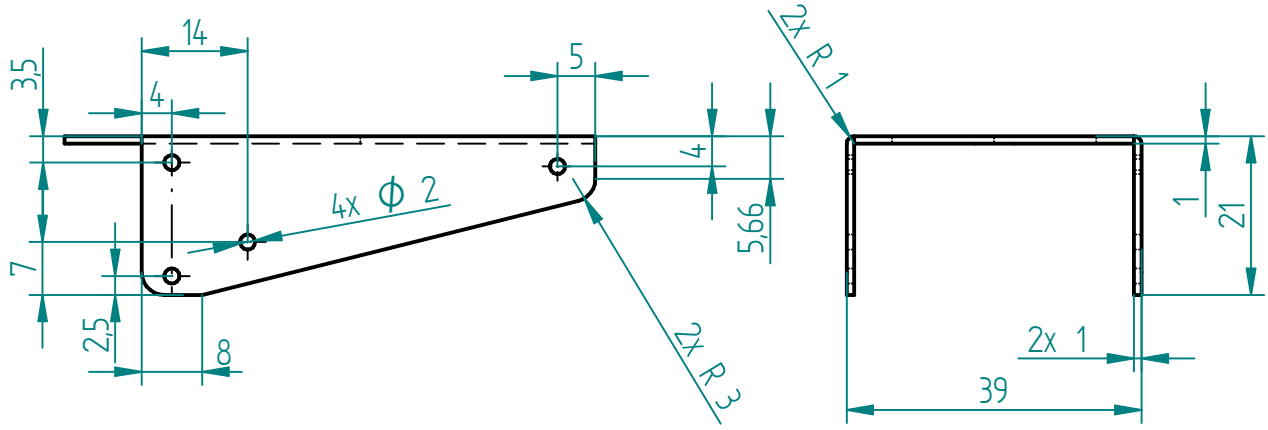
uc3m | Universidad Carlos III de Madrid	**Title:** Chest	
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Created by: Iris Valverde Blanco	**Kind of draft:** Part draft	**Date:** 10/12/2023
		Page number:

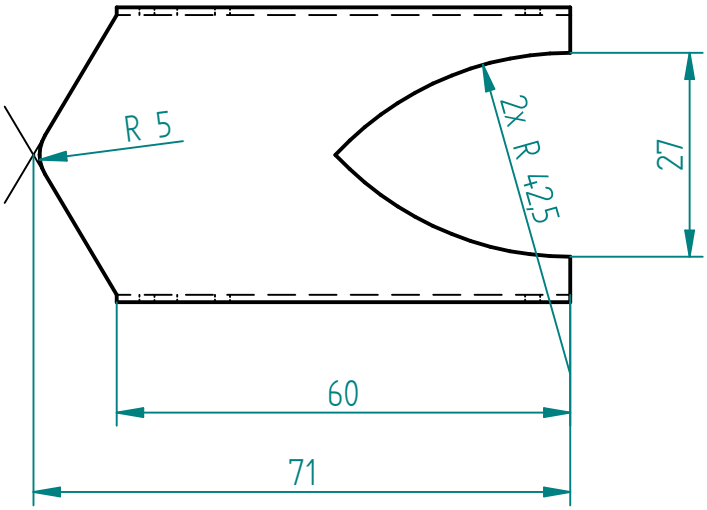




uc3m | Universidad Carlos III de Madrid	**Title:** Hand	
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Created by: [redacted]		**Date:** 26/11/2023
	Kind of draft: Part draft	**Page number:**

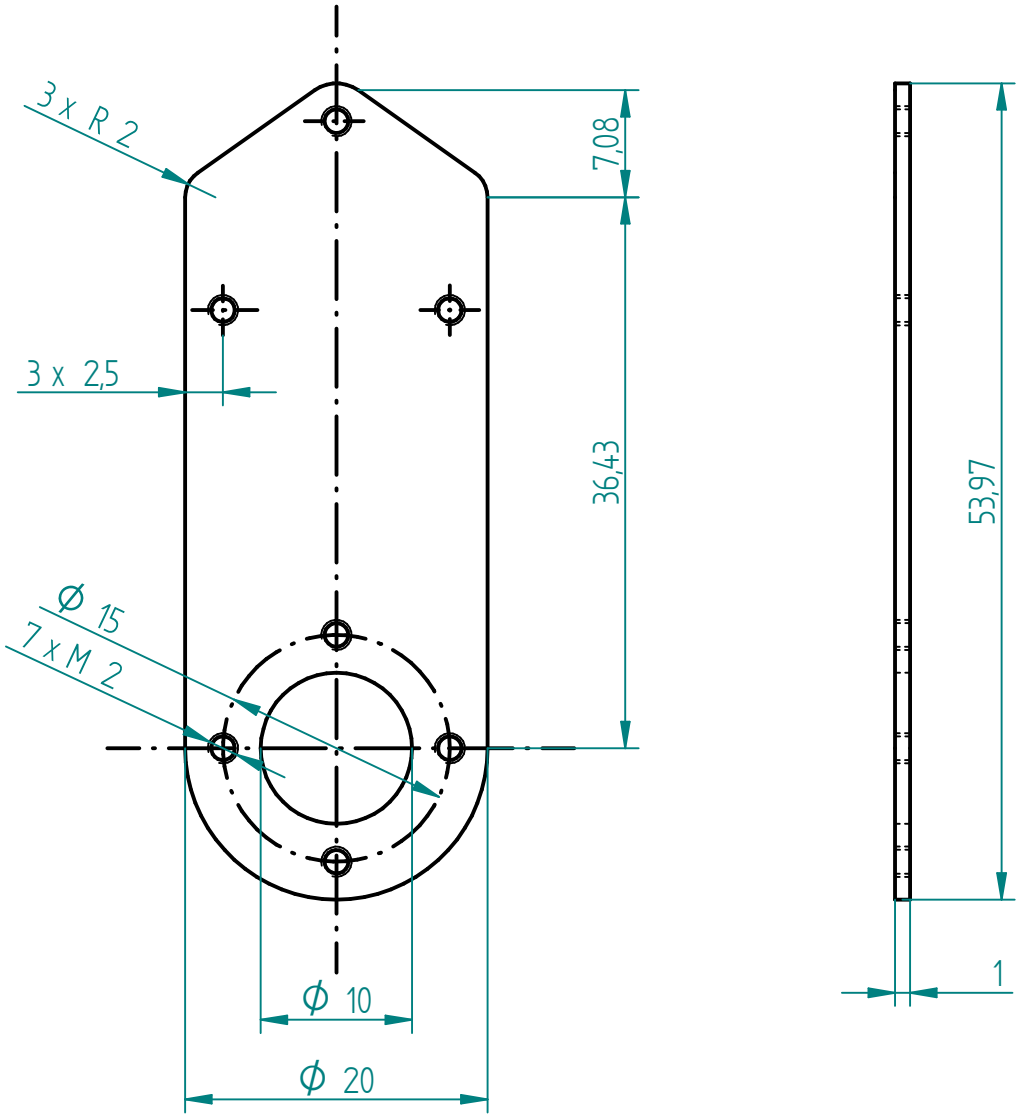
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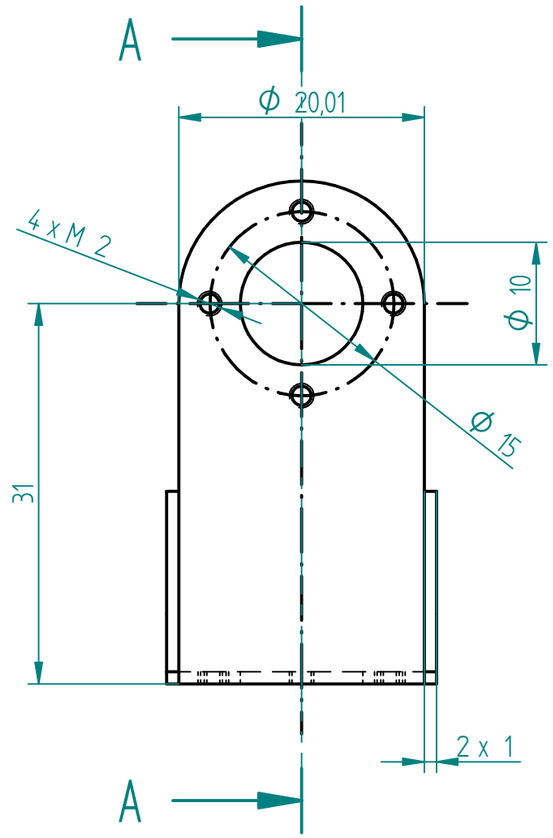
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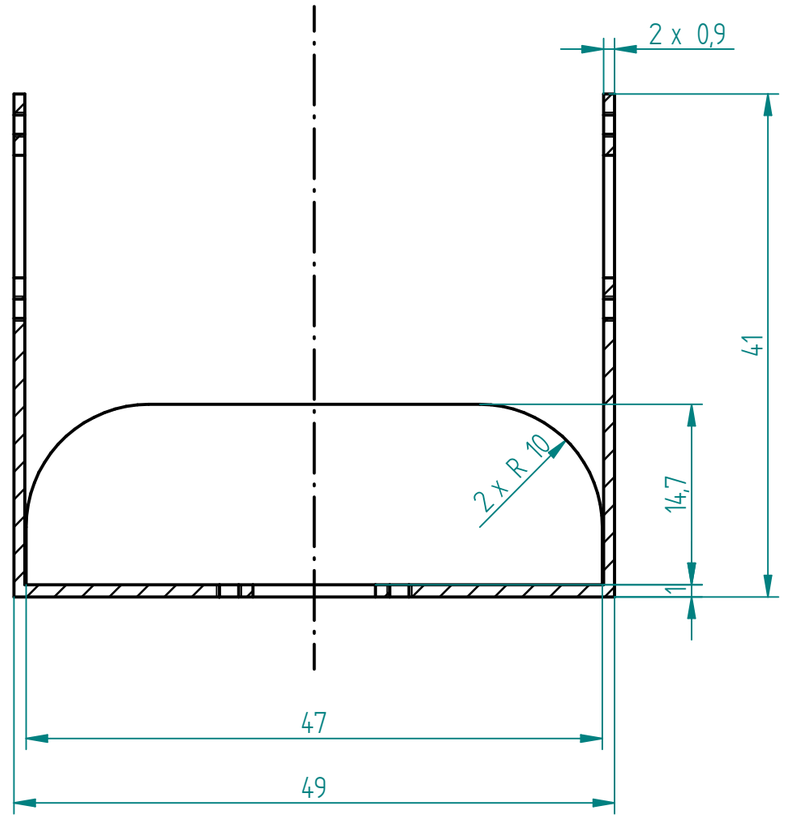
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Created by: [redacted]		**Date:** 15/11/23
	Kind of draft:	**Page number:**

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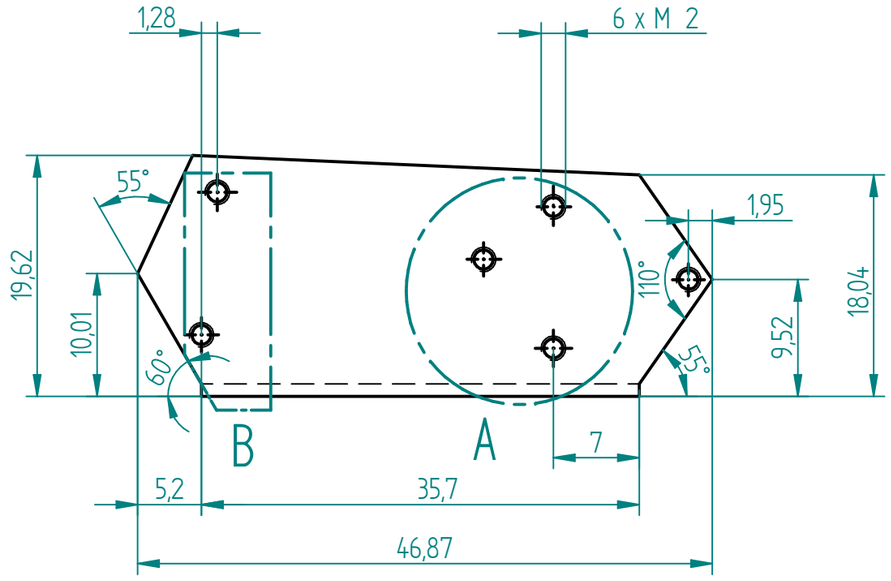


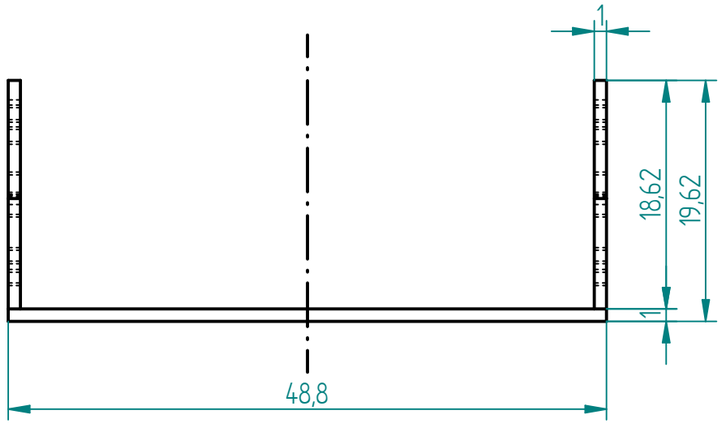


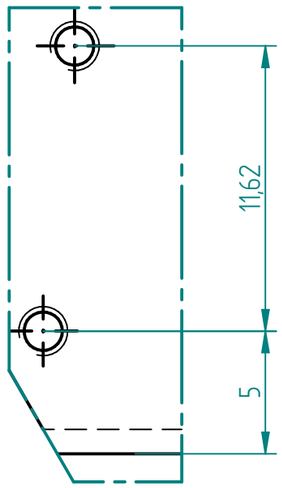
CORTE A-A

SOLID EDGE ACADEMIC COPY

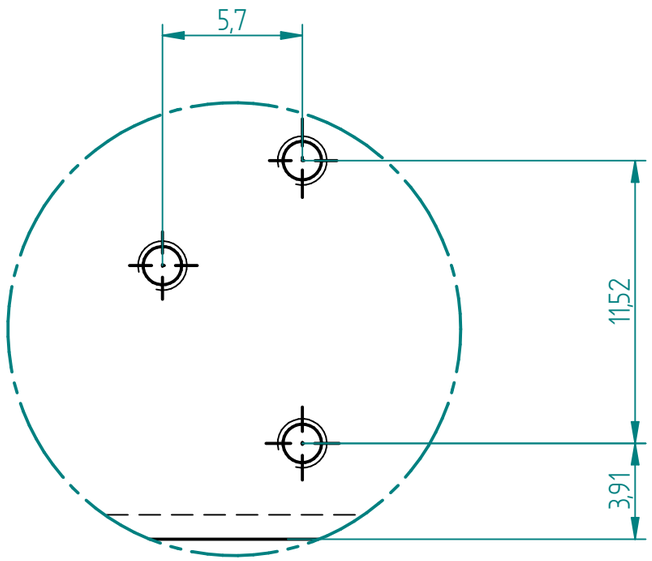
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Created by: [redacted]	**Kind of draft:**	**Page number:**







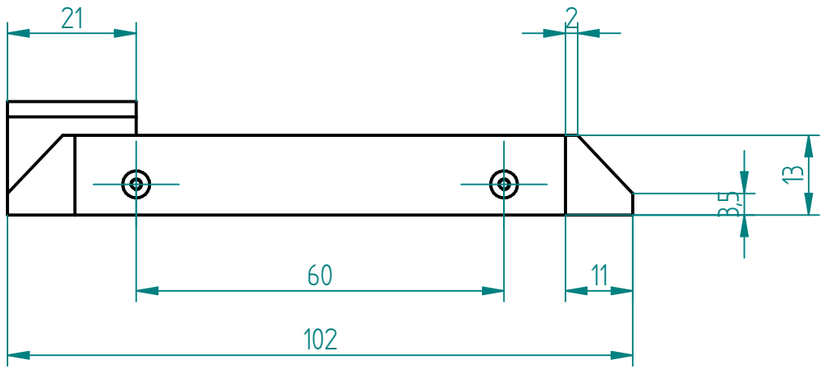
DETALLE B

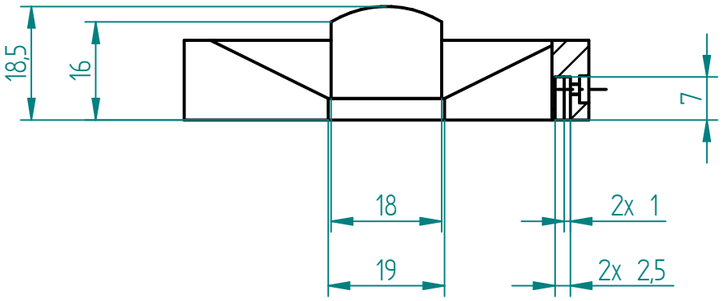


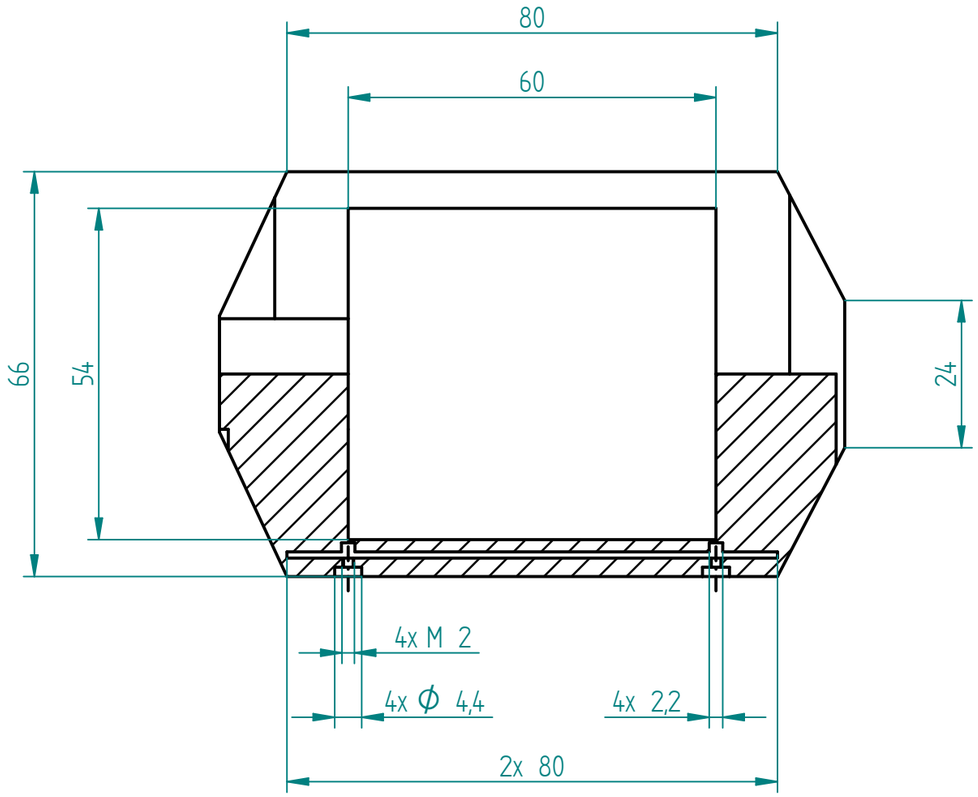
DETALLE A

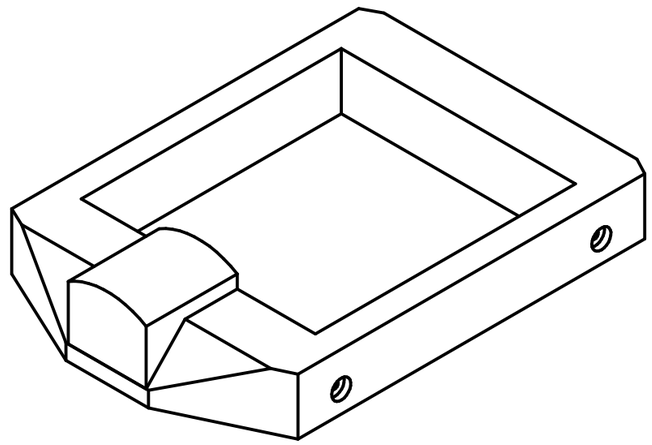
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uc3m | Universidad Carlos III de Madrid	**Title:** Double arrow	
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Created by: [redacted]		**Date:** 03/11/23
	Kind of draft:	**Page number:**



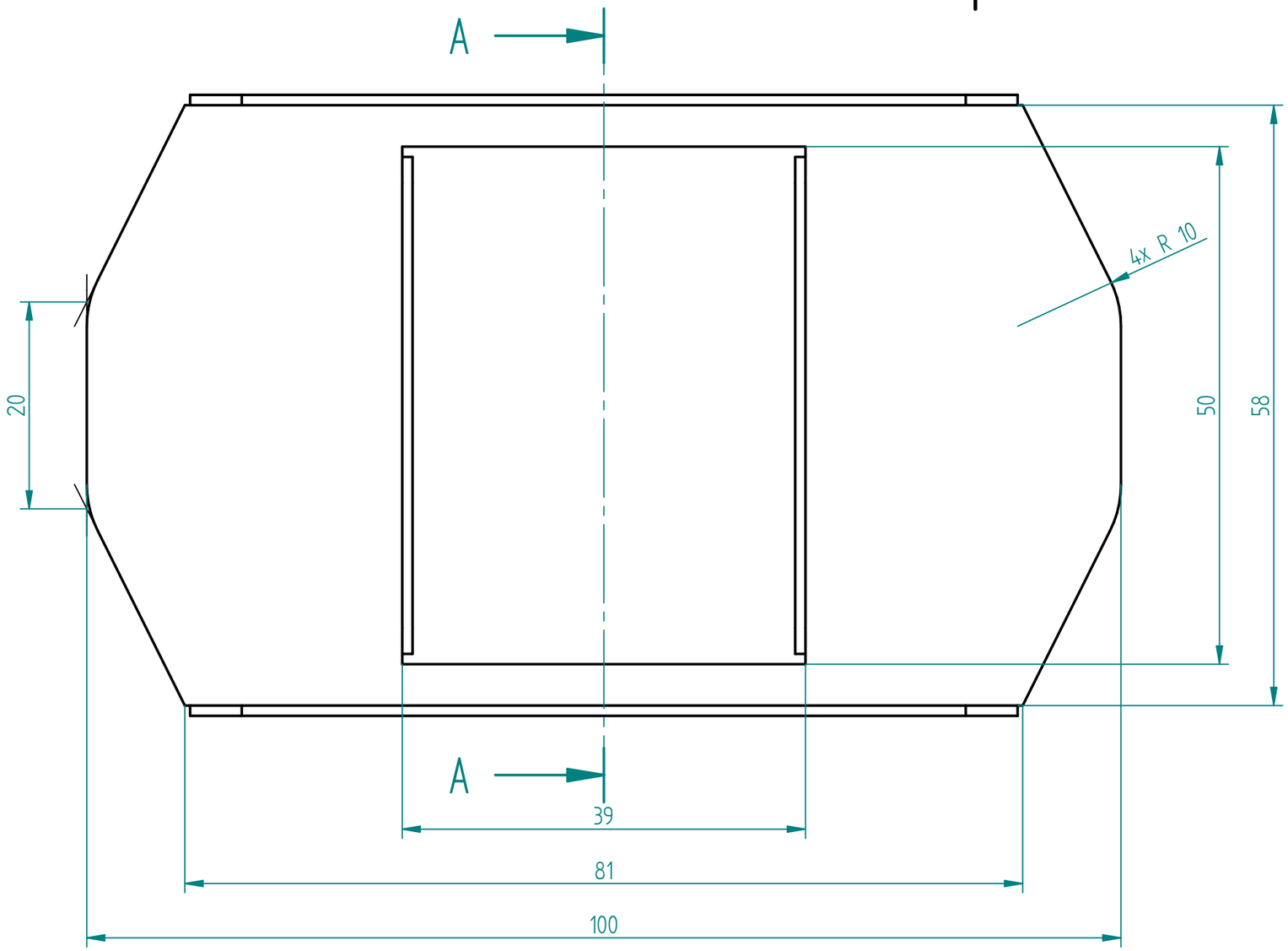


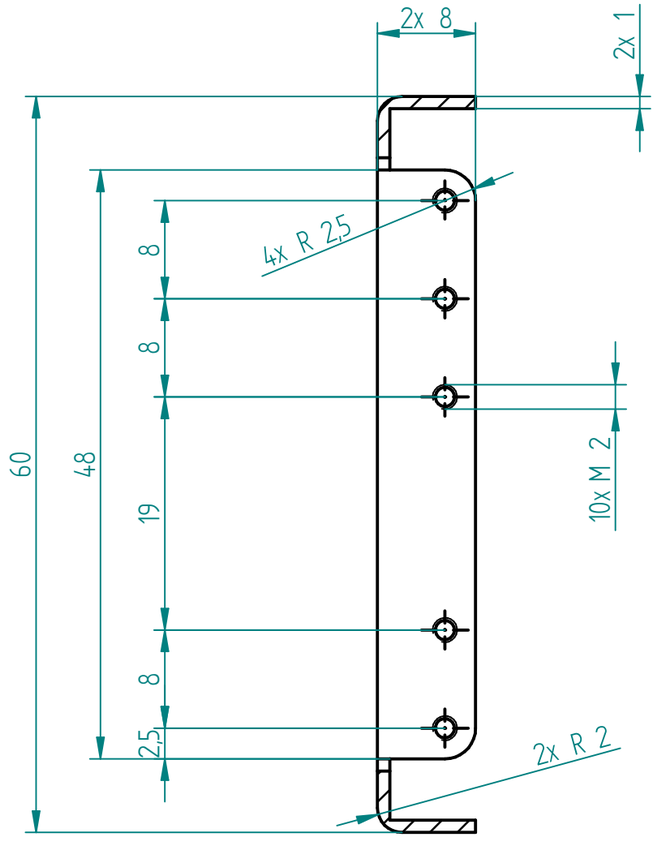




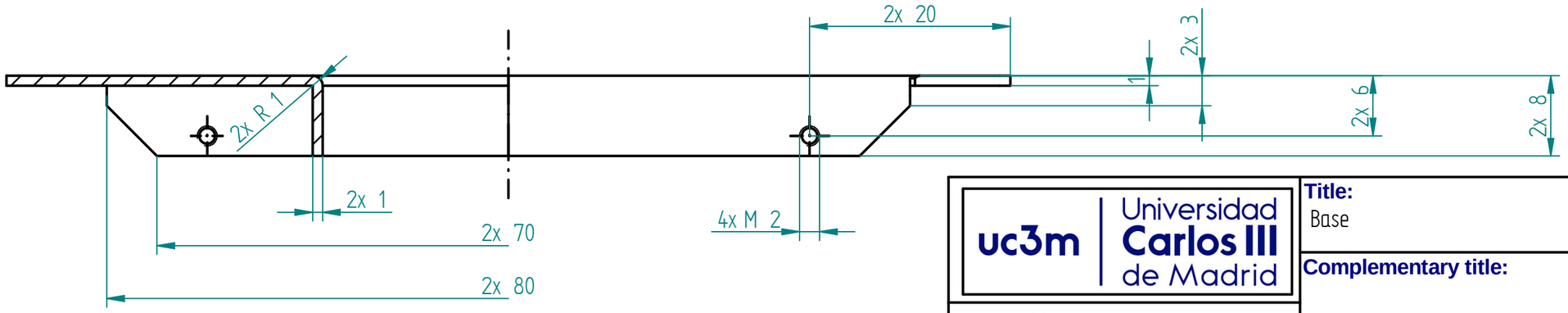
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Created by: Iris Valverde Blanco		**Date:** 06/12/2023
	Kind of draft: Part draft	**Page number:**

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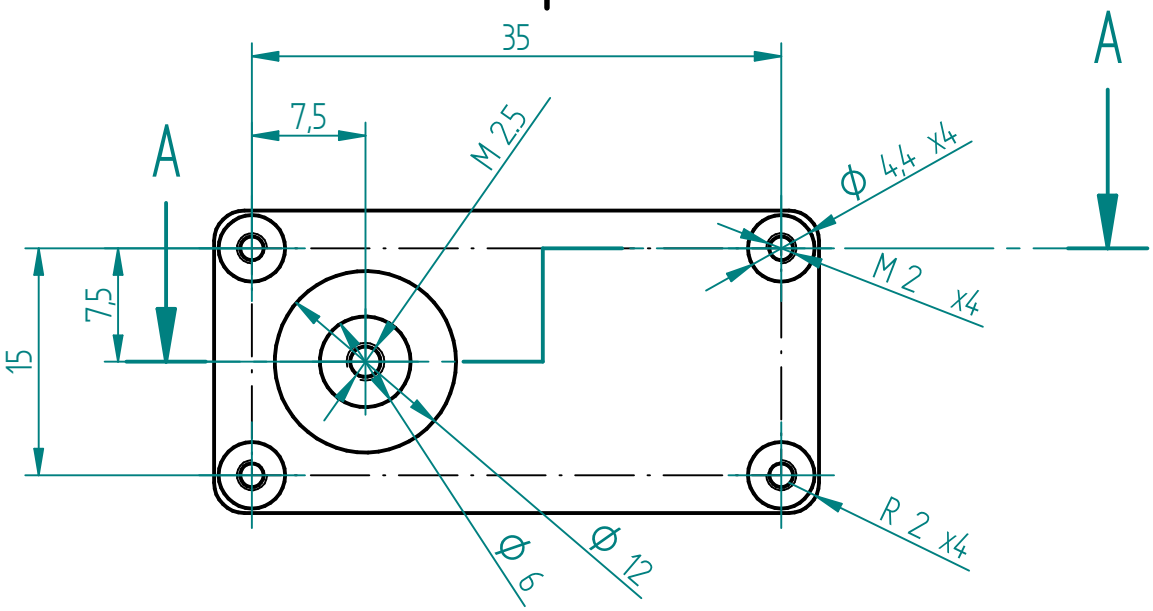


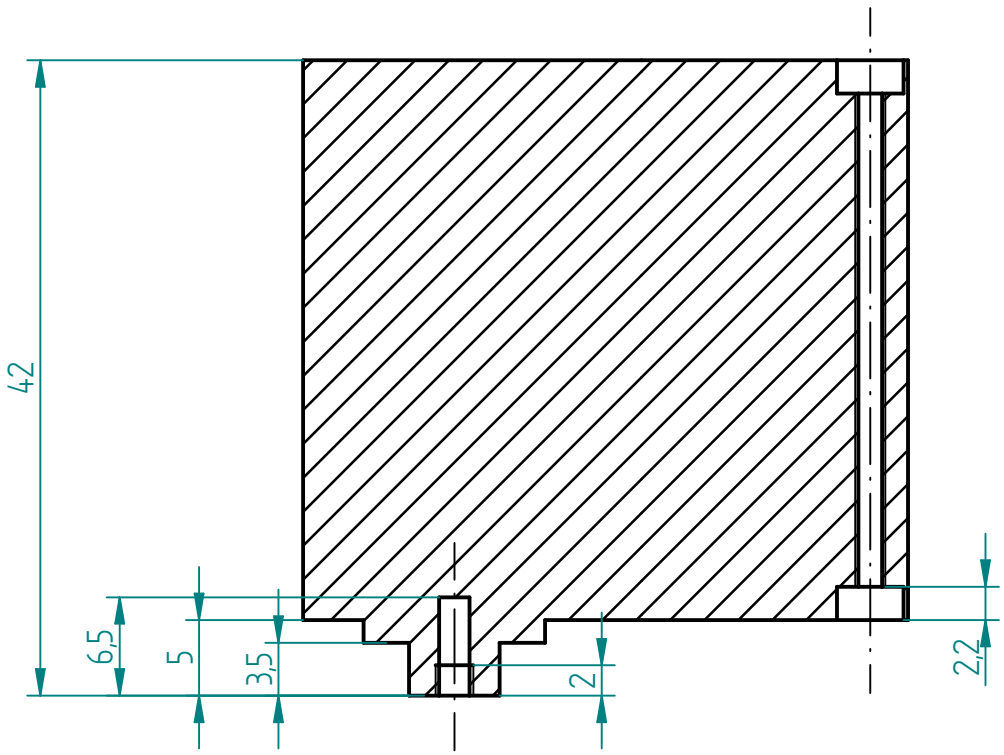
CORTE A-A



SOLID EDGE ACADEMIC COPY

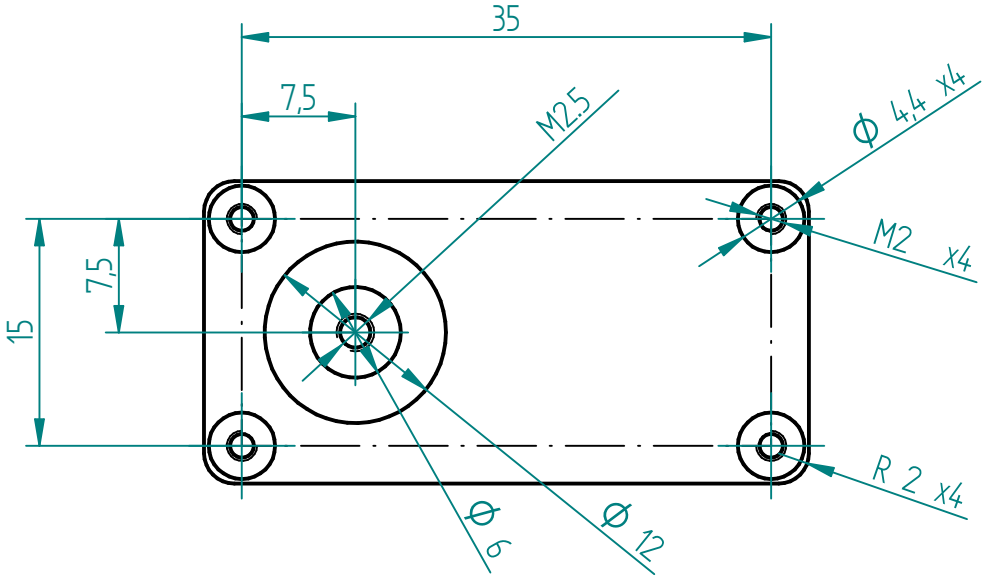
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Created by: Iris Valverde Blanco	**Kind of draft:** Part draft	**Page number:**

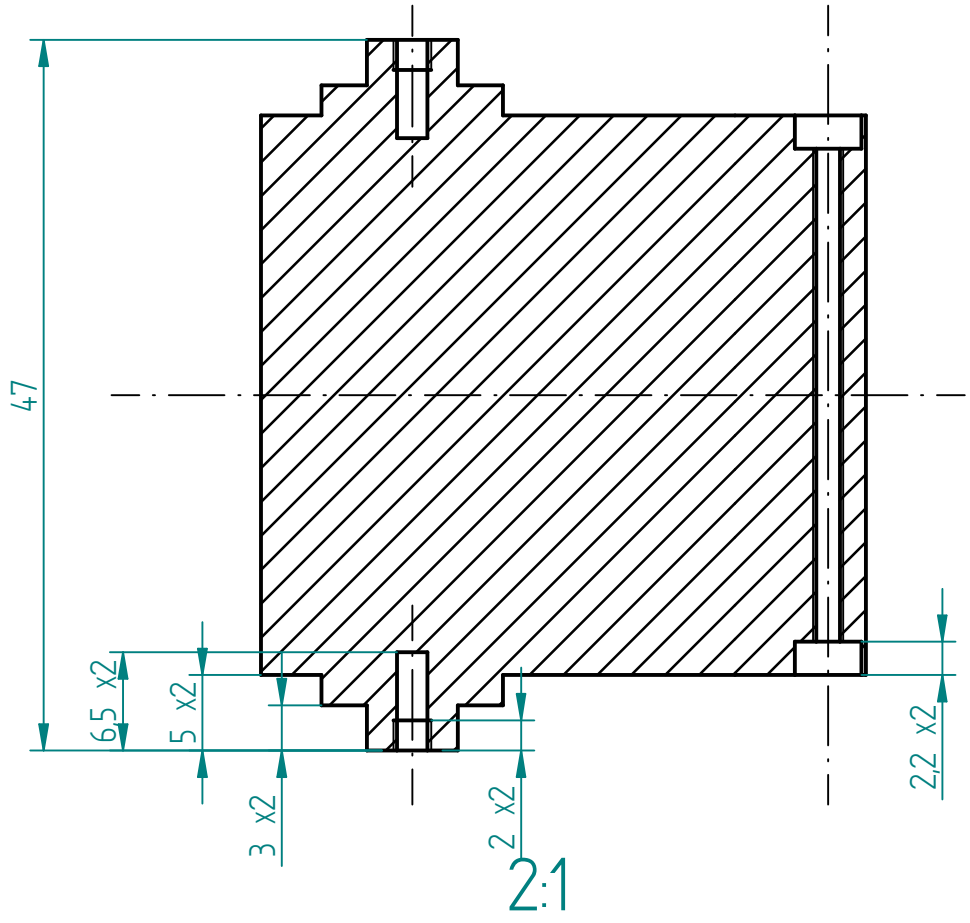




uc3m | Universidad Carlos III de Madrid	**Title:** Single motor	
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Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 7/12/23
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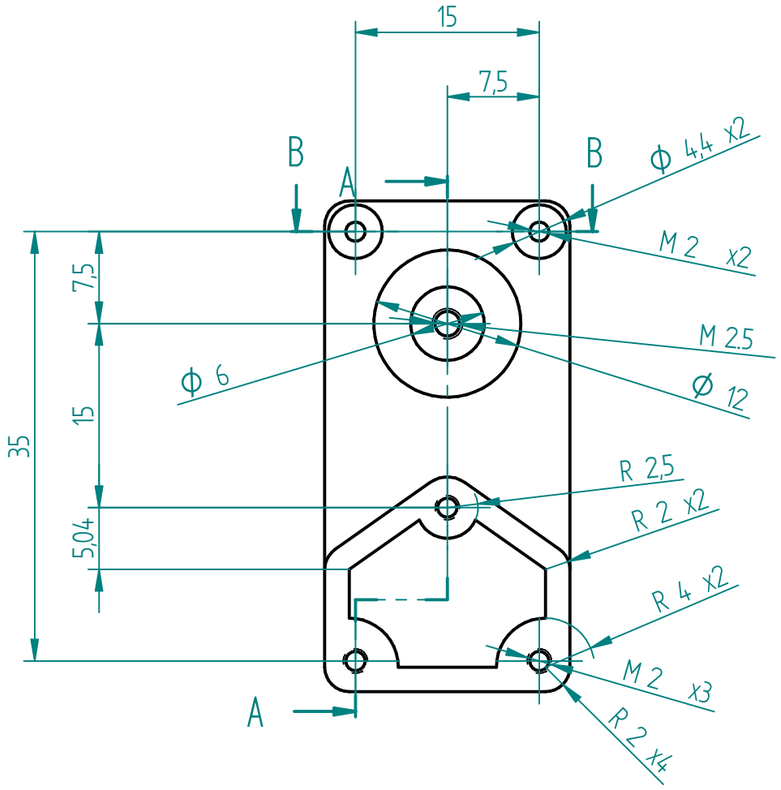
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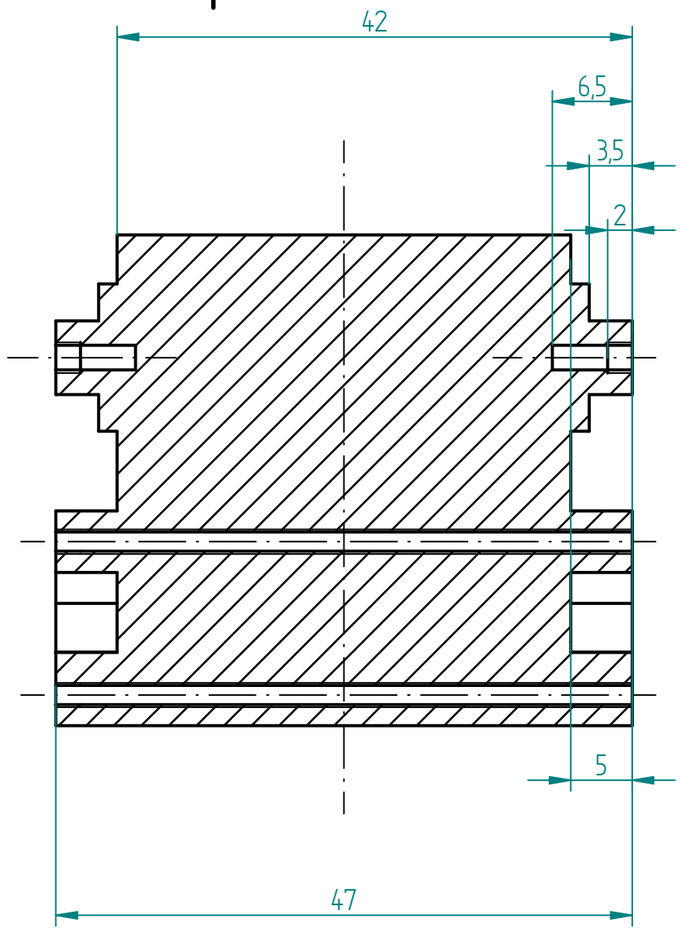


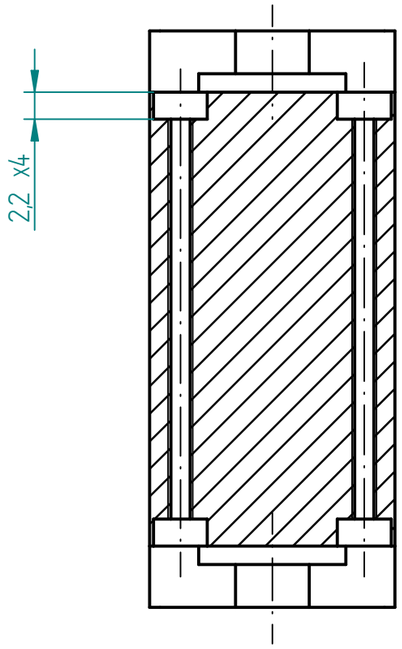


uc3m | Universidad Carlos III de Madrid	**Title:** Double motor	
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Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 7/12/23
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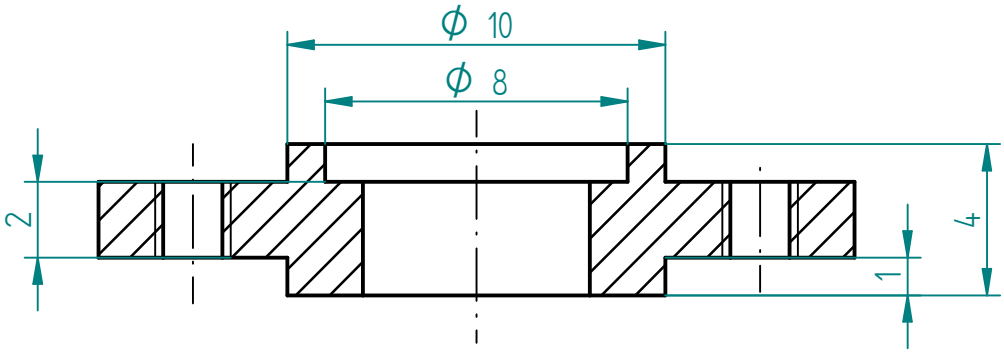


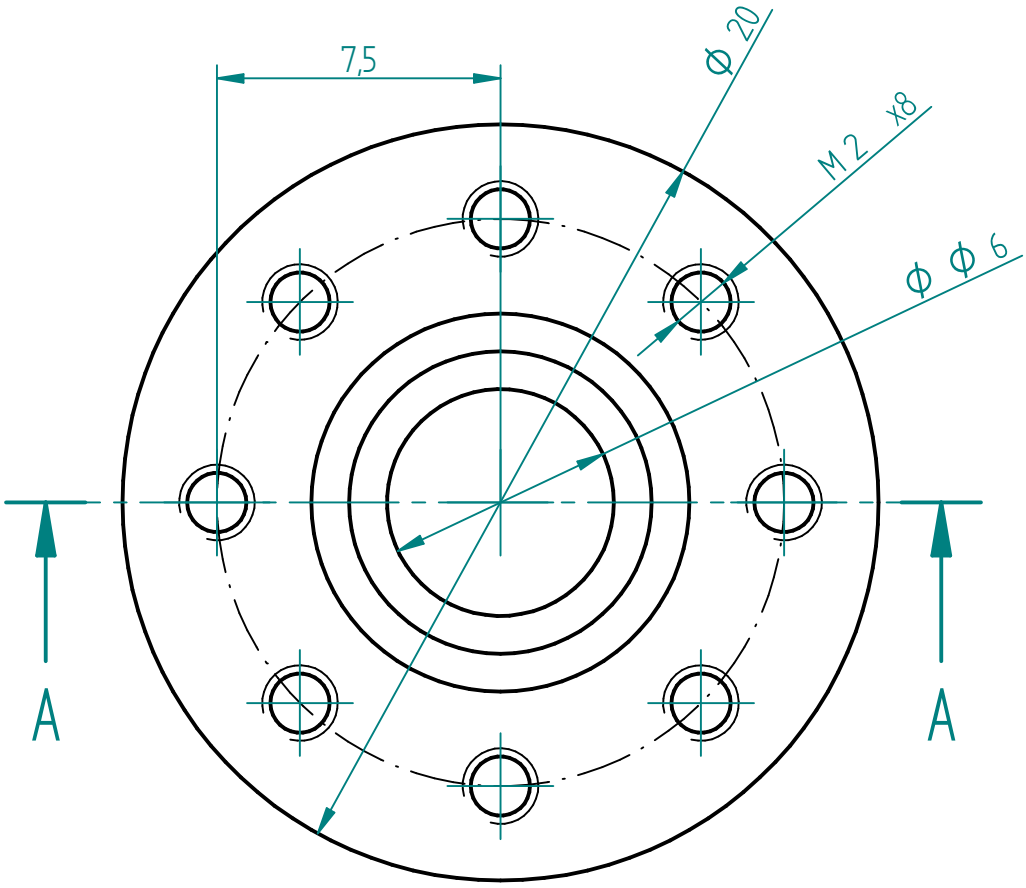




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uc3m | Universidad Carlos III de Madrid	**Title:** Motor with protrusions	
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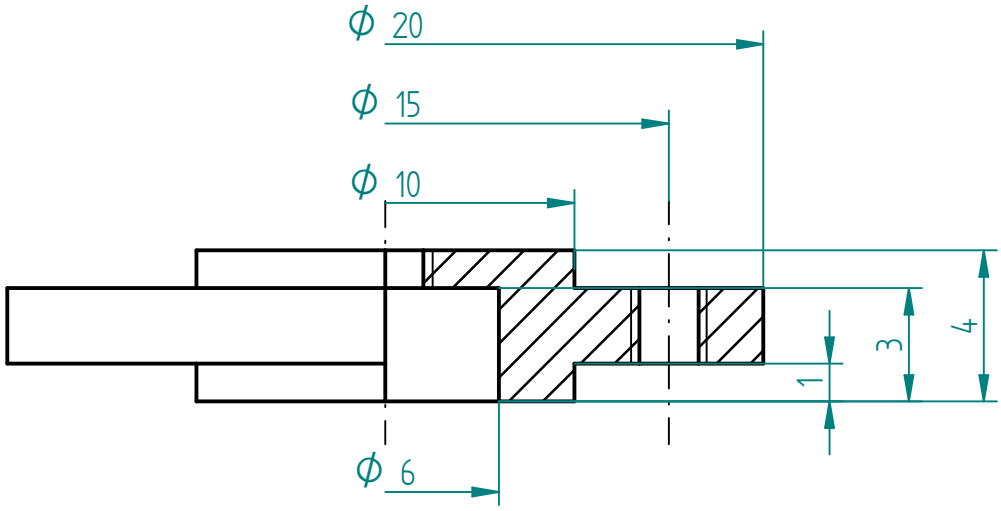


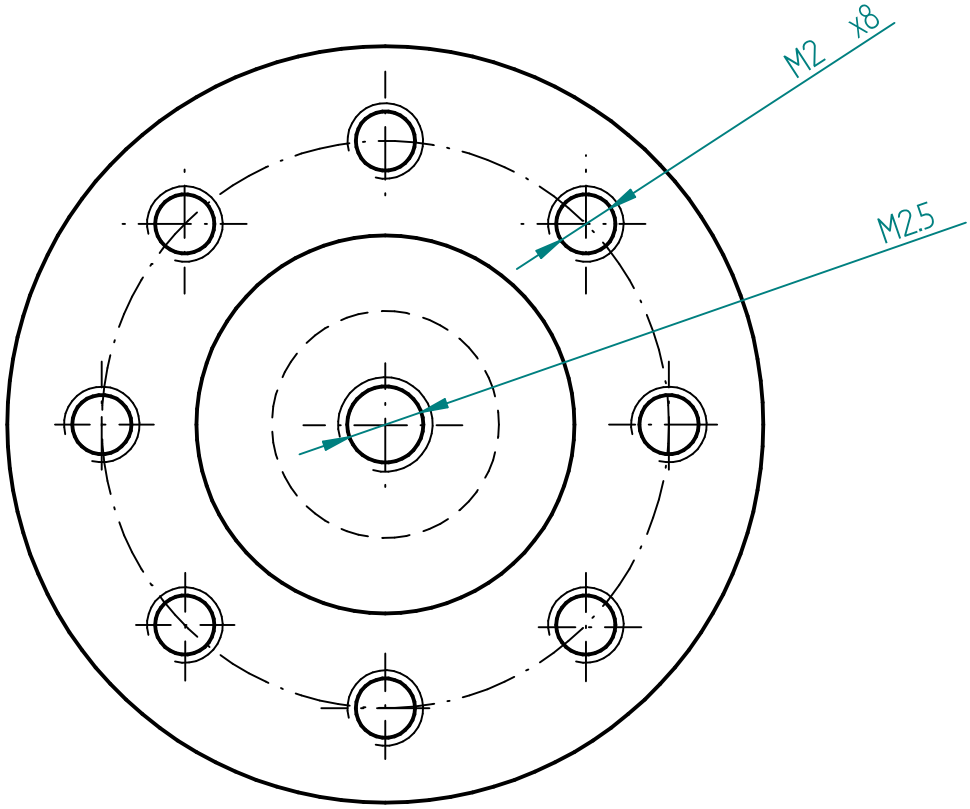


5:1

uc3m | Universidad Carlos III de Madrid	**Title:** Simple horn	
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Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 28/11/2023
		Page number:

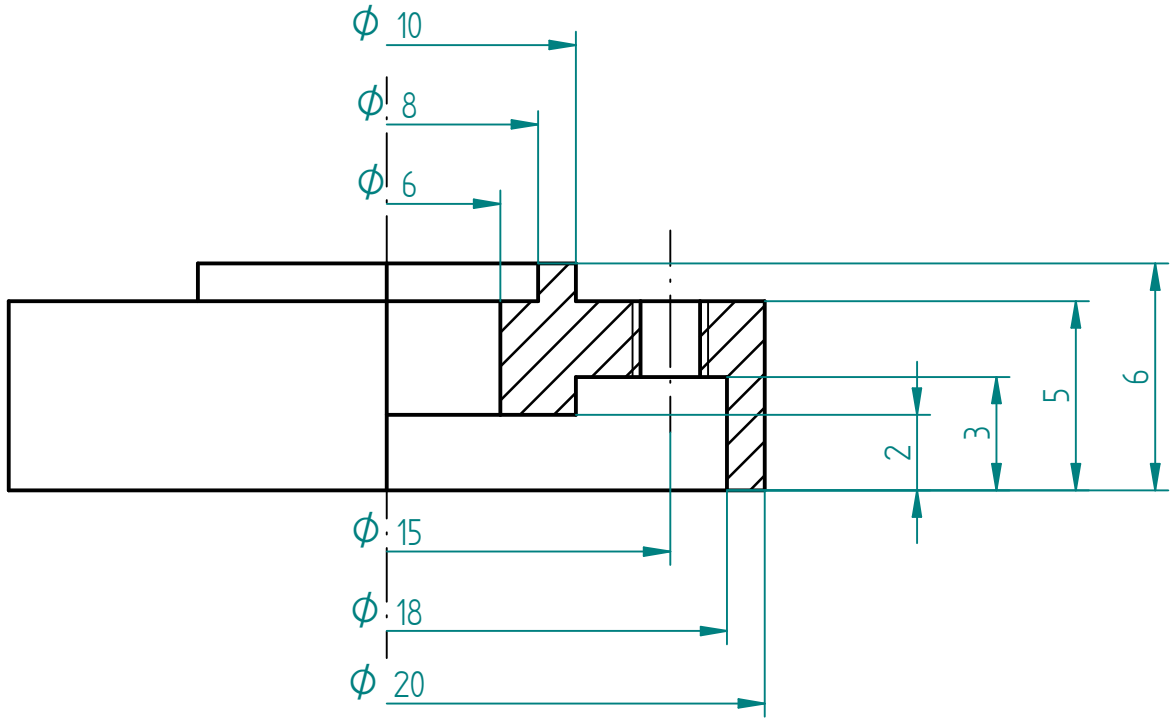
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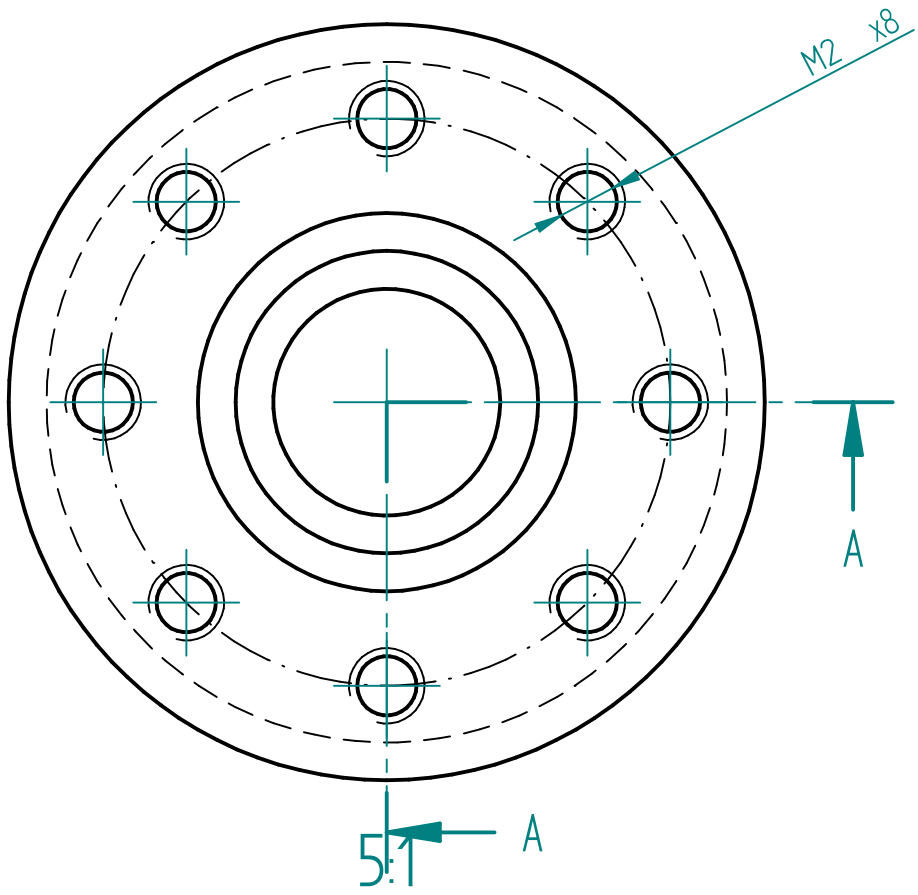




uc3m | Universidad Carlos III de Madrid	**Title:** Simple axis horn	
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Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 28/11/2023
		Page number:

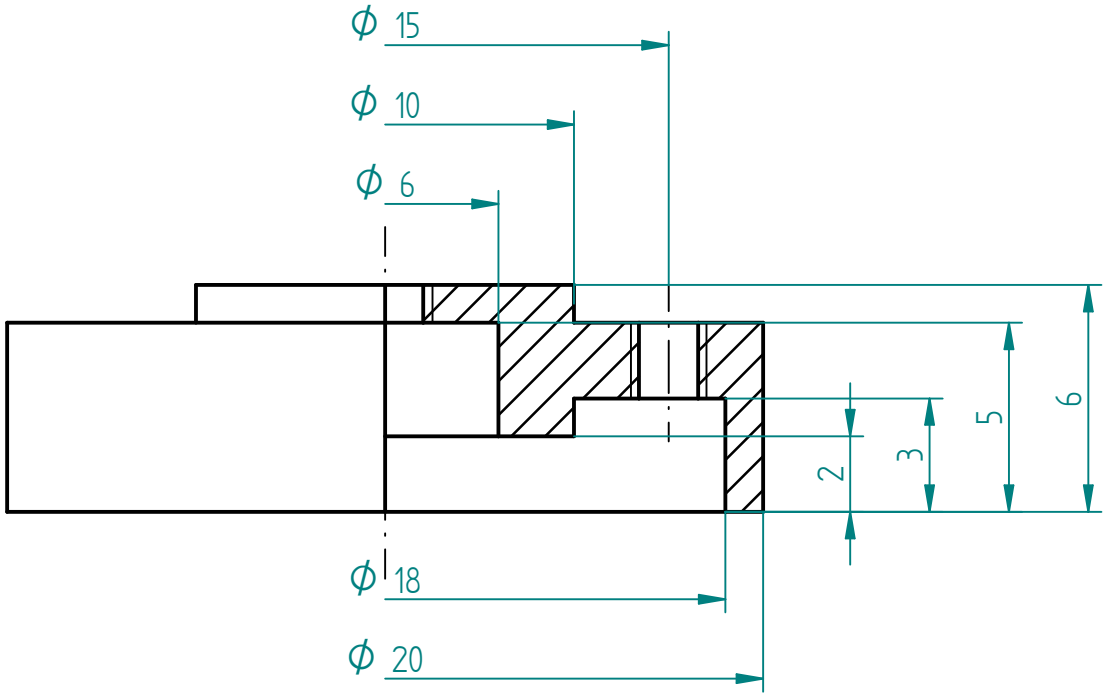
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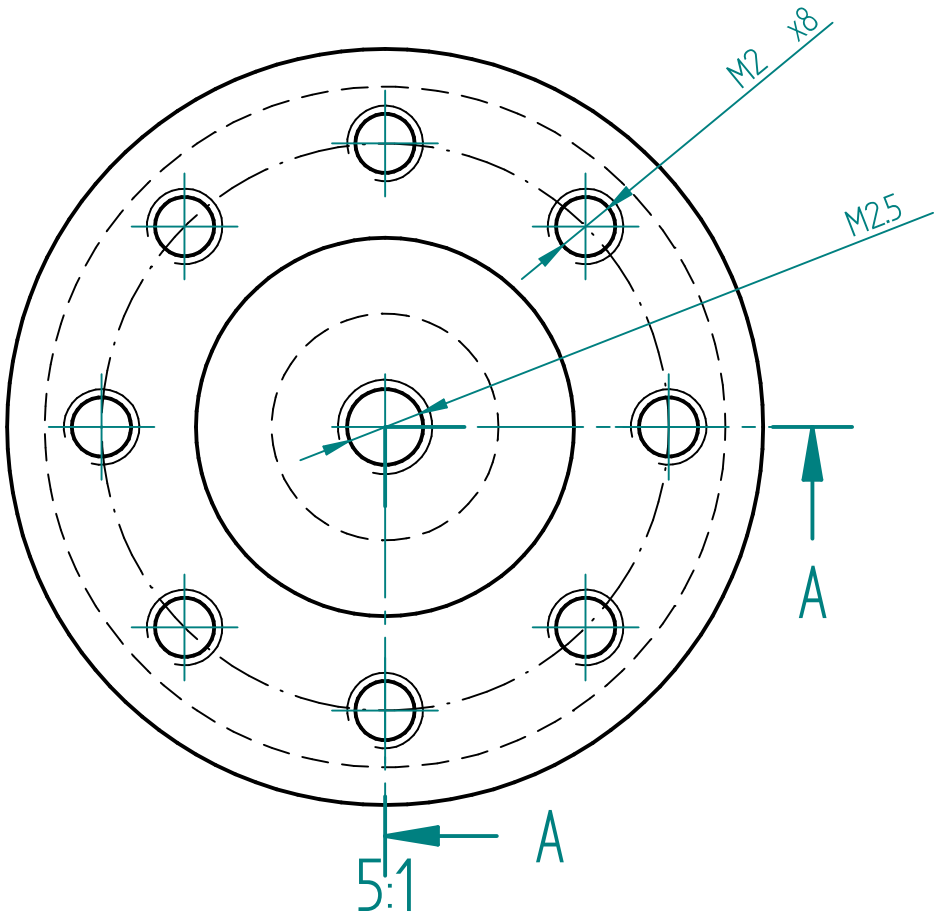




uc3m | Universidad Carlos III de Madrid	**Title:** Thick horn	
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Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 28/11/2023
		Page number:

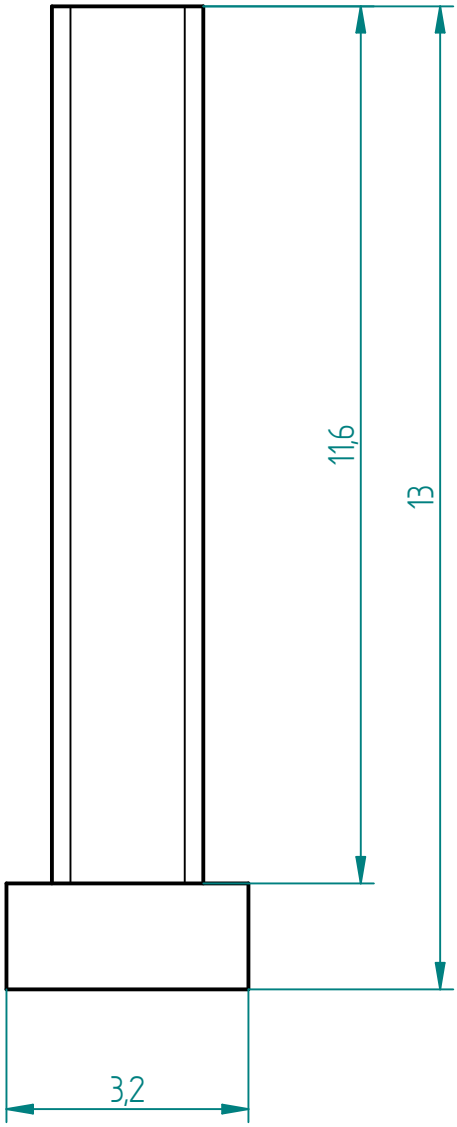
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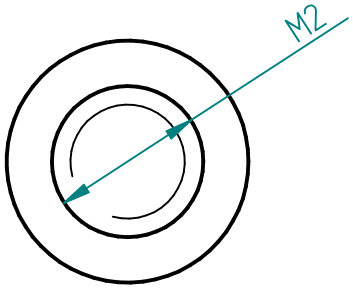




uc3m | Universidad Carlos III de Madrid	**Title:** Thick horn axis	
	Complementary title:	**Scale:** 5:1
Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 28/11/2023
		Page number:

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10:1

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Created by: [redacted]

Title: Screw 1

Complementary title:

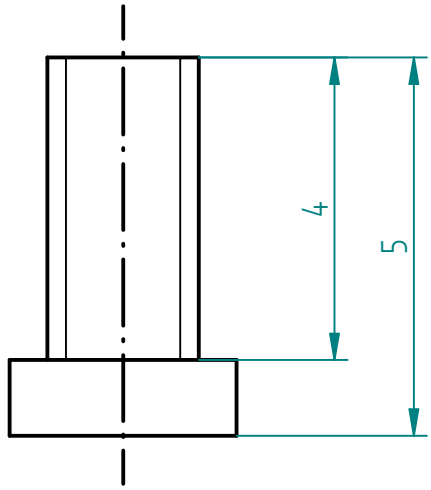
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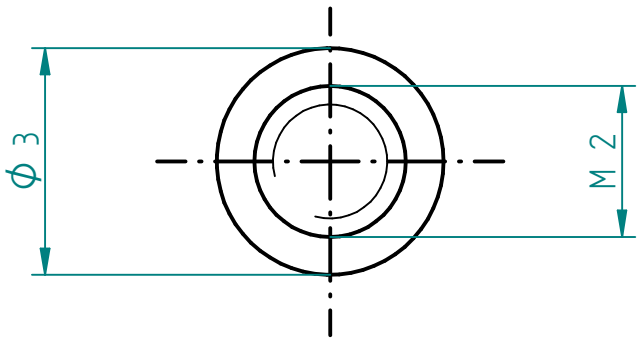
Scale: 10:1

Date: 28/11/2023

Page number:

SOLID EDGE ACADEMIC COPY





uc3m | Universidad Carlos III de Madrid

Created by:
Iris Valverde Blanco

Title:
Screw2

Complementary title:

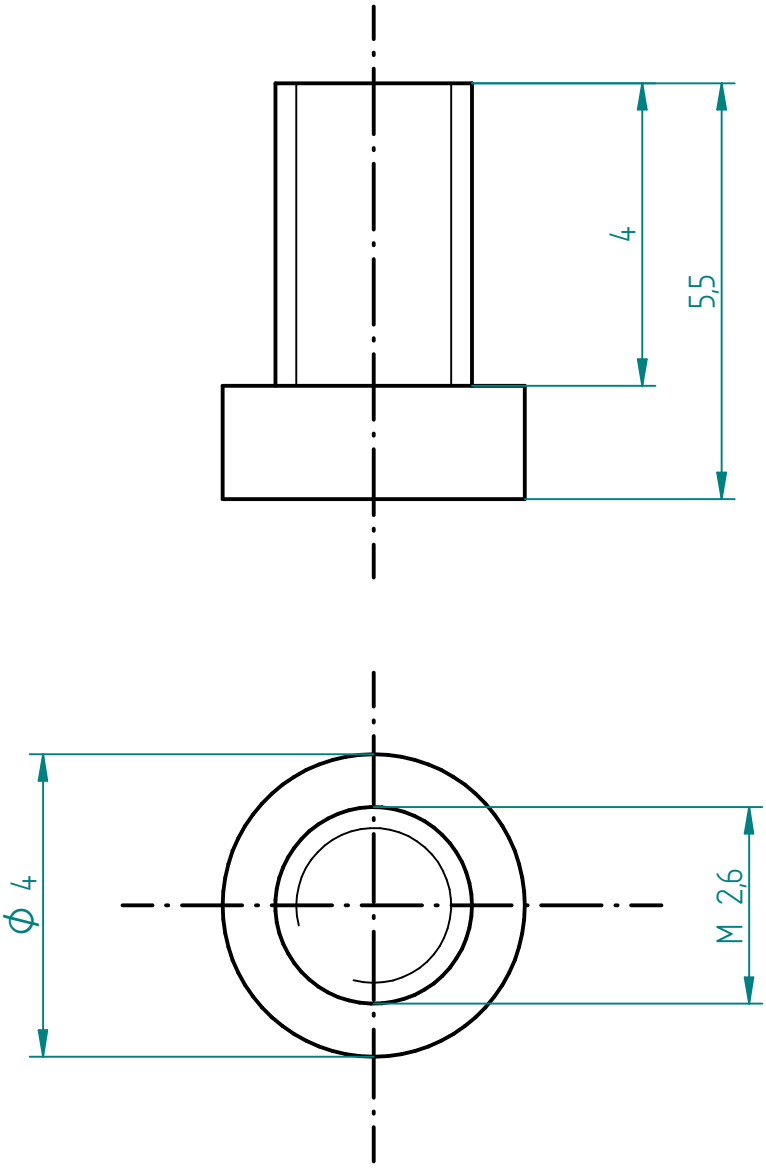
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Scale:
10:1

Date:
02/12/2023

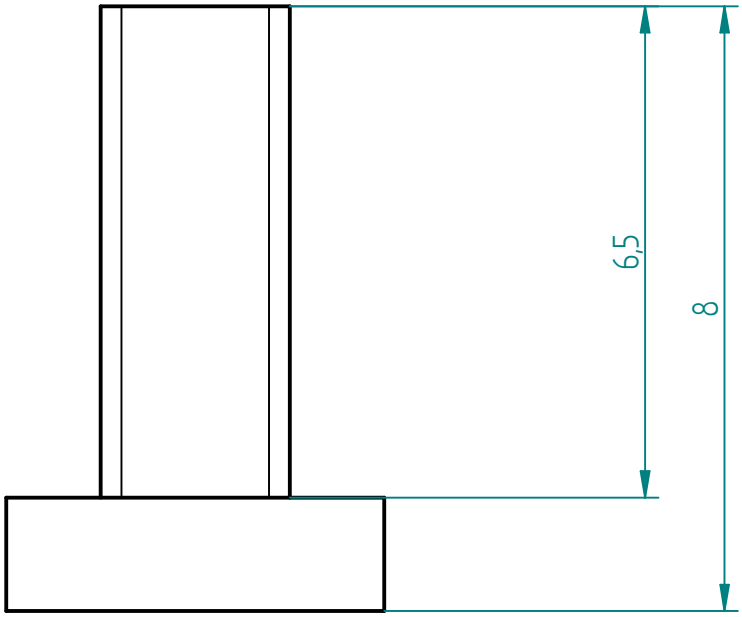
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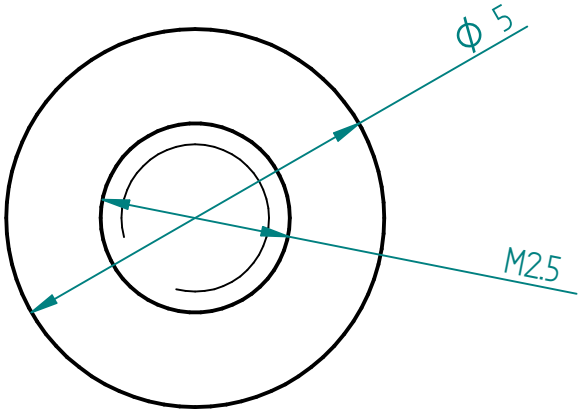
SOLID EDGE ACADEMIC COPY



uc3m | Universidad Carlos III de Madrid	**Title:** Screw3	
	Complementary title:	**Scale:** 10:1
Created by: Iris Valverde Blanco		**Date:** 02/12/2023
	Kind of draft: Part draft	**Page number:**

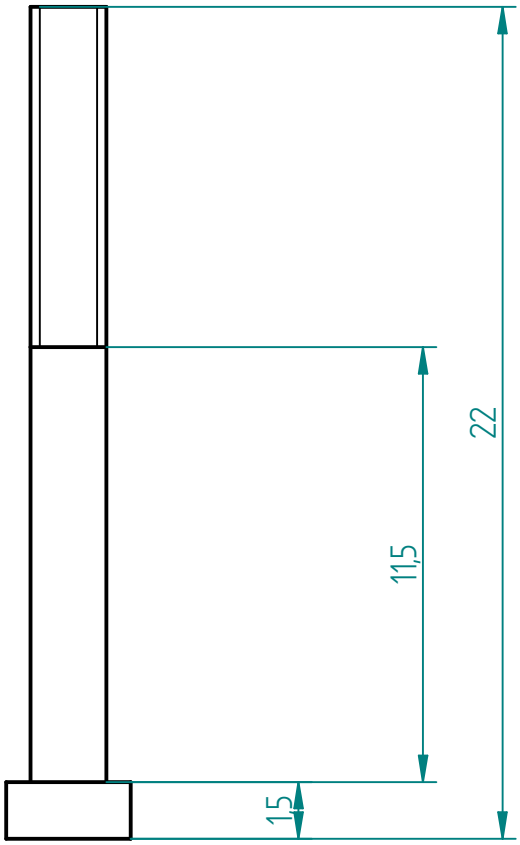
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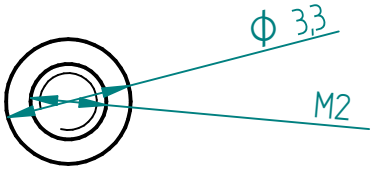




uc3m | Universidad Carlos III de Madrid	**Title:** Screw motor axis	
	Complementary title: Screw for the motor axis	**Scale:** 5:1
Created by: [redacted]	**Kind of draft:** Part draft	**Date:** 28/11/2023
		Page number:

SOLID EDGE ACADEMIC COPY





5:1

uc3m | Universidad Carlos III de Madrid

Created by: [redacted]

Title: Screw 2 motor

Complementary title:

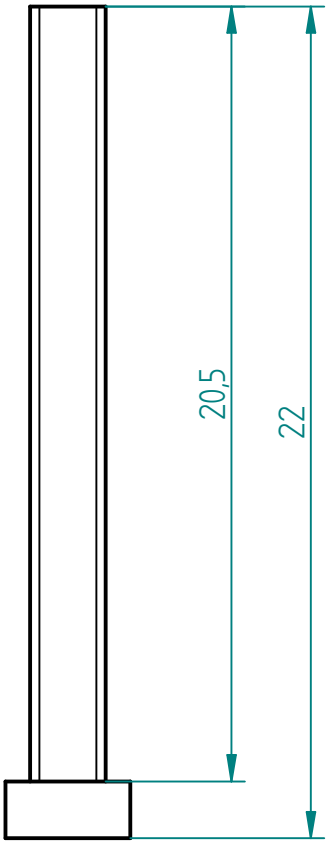
Kind of draft: Part draft

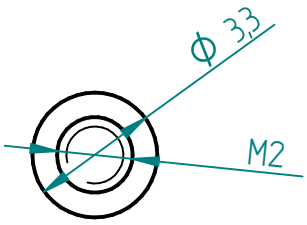
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Date: 28/11/2023

Page number:

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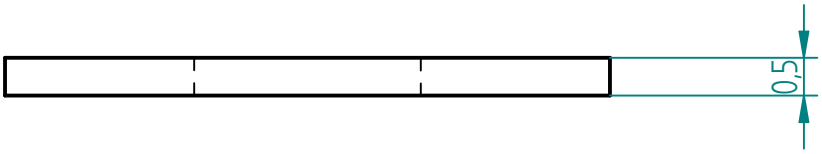


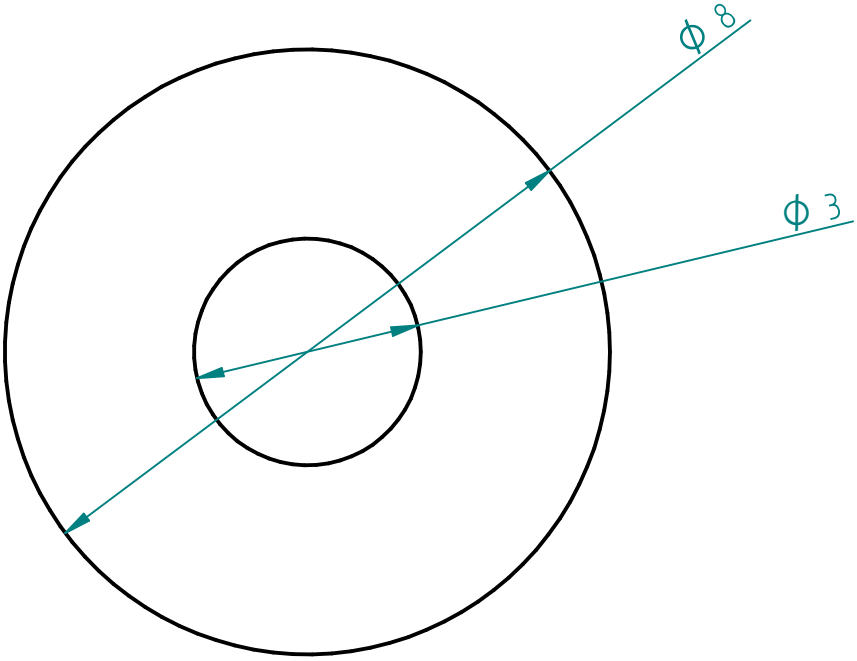


5:1

uc3m | Universidad Carlos III de Madrid	**Title:** Screw 2.2 motor	
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		Page number:

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10:1

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	Complementary title:	**Scale:** 10:1
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